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Experiment 1

Determination of electron specific charge using the Helmholtz coil

1.1 Purpose of the experiment

- Studying the motion of electrons in a uniform magnetic field
- Determining the specific charge of the electron

1.2 Theoretical context of the experiment

When an electron moves at a velocity $\vec{v}$ in a magnetic field of induction $\vec{B}$, it experiences a Lorentz force

$$\vec{F} = (-e)\vec{v} \times \vec{B} \tag{1.1}$$

where $e = 1.6 \times 10^{-19} C$ is the elementary charge. At each moment this force is perpendicular to the direction of the field (vector $\vec{B}$) and to the direction of motion ($\vec{v}$). Forces which are perpendicular to the direction of motion cannot change the magnitude of the velocity since their work is zero. However, they will change the direction of motion, i.e. the direction of $\vec{v}$.

Let us consider that the magnetic field does not change in time (we say that the field is uniform). We can always write the velocity $\vec{v}$ as a superposition of two components, one which is perpendicular ($\vec{v_\perp}$) and another which is parallel ($\vec{v_\parallel}$) to the magnetic field: $\vec{v} = \vec{v_\perp} + \vec{v_\parallel}$. Lorentz force changes only the direction of $\vec{v_\perp}$, by acting on the electron as a centripetal force which produces a circular motion in the perpendicular plane to the field. On the other hand,$\vec{v_\parallel}$ stays constant, which means that the electron is translated in the direction of the field at this component of the velocity. Consequently, the resultant trajectory of an electron in a uniform magnetic field is in general a spiral.

If the electron is entering the field at a right angle ($\vec{v} = \vec{v_\perp}$ then the Lorentz force produced by the magnetic field $F = ev_\perp B = evB$ sets the electron on a circular trajectory of radius r :

$$F = evB = \frac{mv^2}{r} \rightarrow r = \frac{mv}{eB}. \tag{1.2}$$

Then the specific charge of the electron is

$$\frac{e}{m} = \frac{v}{rB} \tag{1.3}$$

To obtain an experimental value of the specific charge of the electron, one has to create the conditions for measuring the trajectory radius of electrons with well determined velocity in a well determined, uniform, magnetic field. It is a type of experiment that was performed for the first time by J.J. Thomson, in 1897.

1.2.1 Accelerating electrons in an electric field

When an electron considered to be at rest is accelerated under a difference of potential U, it will acquire a kinetic energy which is equal to the work performed by the electric field forces (work-energy theorem)

$$\frac{mv^2}{2} = eU \tag{1.4}$$

According to eq. 1.4, the squared speed of the electron is proportional to the accelerating voltage:

$$v^2 = \frac{2eU}{m} \tag{1.5}$$

1.2.2 Magnetic field produced by the Helmholtz coil

A system composed of two identical coils of radius R separated by a distance equal to R is called Helmholtz coil. It will provide an almost uniform magnetic field in a limited region around the axis of the coils, when the coils are run by the same current of intensity I, in the same direction. It can be demonstrated by applying Ampere's law, that the dependence of the uniform magnetic field induction B on the electric current I is given by:

$$B = \left(\frac{4}{5}\right)^{3/2} \mu_0 \frac{n}{R} \cdot I \tag{1.6}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the magnetic constant of vacuum and n is the number of turns in a coil.

1.3 Description of the experiment

In this experiment we observe and obtain quantitative information on quasi-monoenergetic electron beams moving in a quasi-uniform magnetic field.

We are going to

- observe and measure the motion of electron beams produced by an electron gun in a uniform magnetic field produced by a pair of Helmholtz coils;
- measure the electron accelerating voltage for which the trajectory in a magnetic field produced by a current in the coils is a circle of a known radius .

The experimental setup is composed of

- pair of Helmholtz coils
- evacuated clear glass tube containing low pressure neon gas (4×10^3 mbar)
- electron gun (inside the tube)
- ladder-shaped measuring device (inside the bulb)

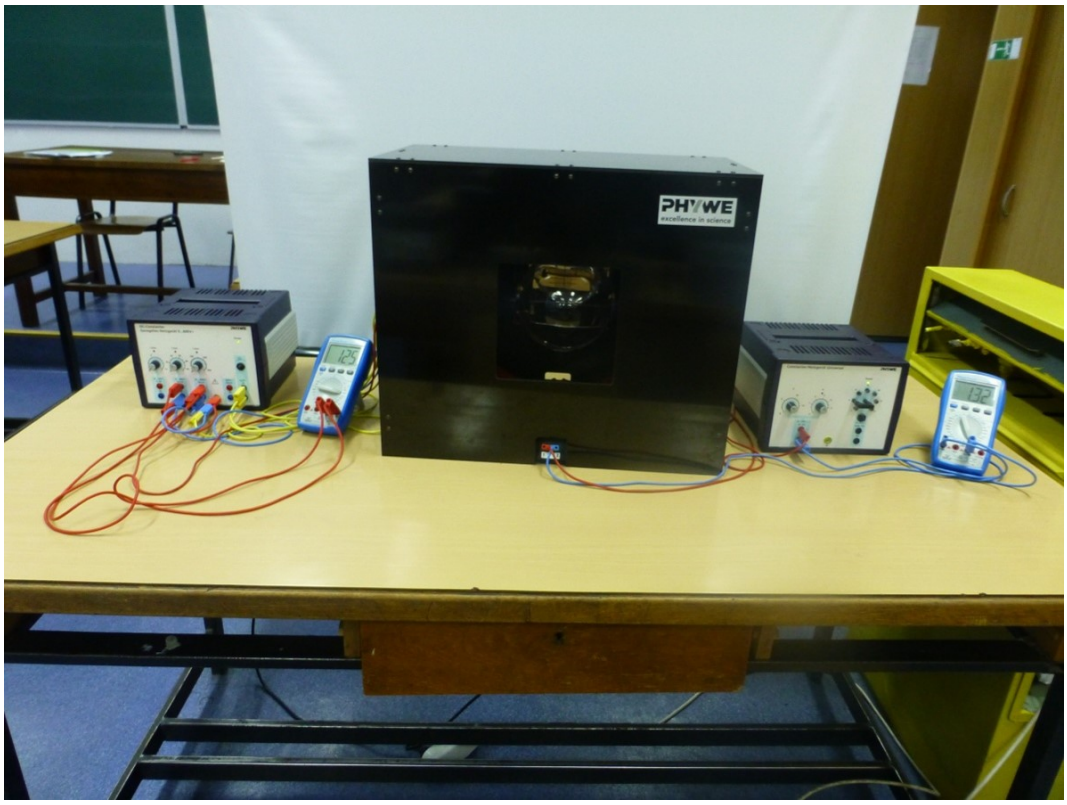


Figure 1.1: Photograph of the experimental setup

- power supply for the filament (6.3V/2A), the selection and focusing grid ($0 \div 50V$) and to accelerating cathode/anode electrodes ($0 \div 300V$)
- voltmeter to measure the accelerating voltage (600 V scale) - see left picture
- power supply for the Helmholtz coils (max 18 V/5A) used at a power of 2W
- ammeter to measure the current intensity through the coils (10 A scale) - see right picture.

The collimated electron beam is produced by an electron gun inside the tube. It is composed of:

- filament - it is heated if an electric current flows through it; the heat necessary for the cathode to release electrons is provided by it;
- grid - its electric potential can be adjusted in order to select and focus the electrons emitted by the cathode.
- cathode - it has a metallic oxide surface which emits electrons by thermionic effect;
- cathode (C) / anode (A) system which is polarized by applying accelerating voltage U; electrons are accelerated in the electric field between C and A;

The electron beam is emitted through a small hole in the anode. With a correct orientation of the tube (mind that it can be rotated in the socket), it is directed in a vertical, upward direction. In the absence of the magnetic field, the beam appears as a straight thick line. Note that what we see is in fact visible light emitted by the host neon atoms when

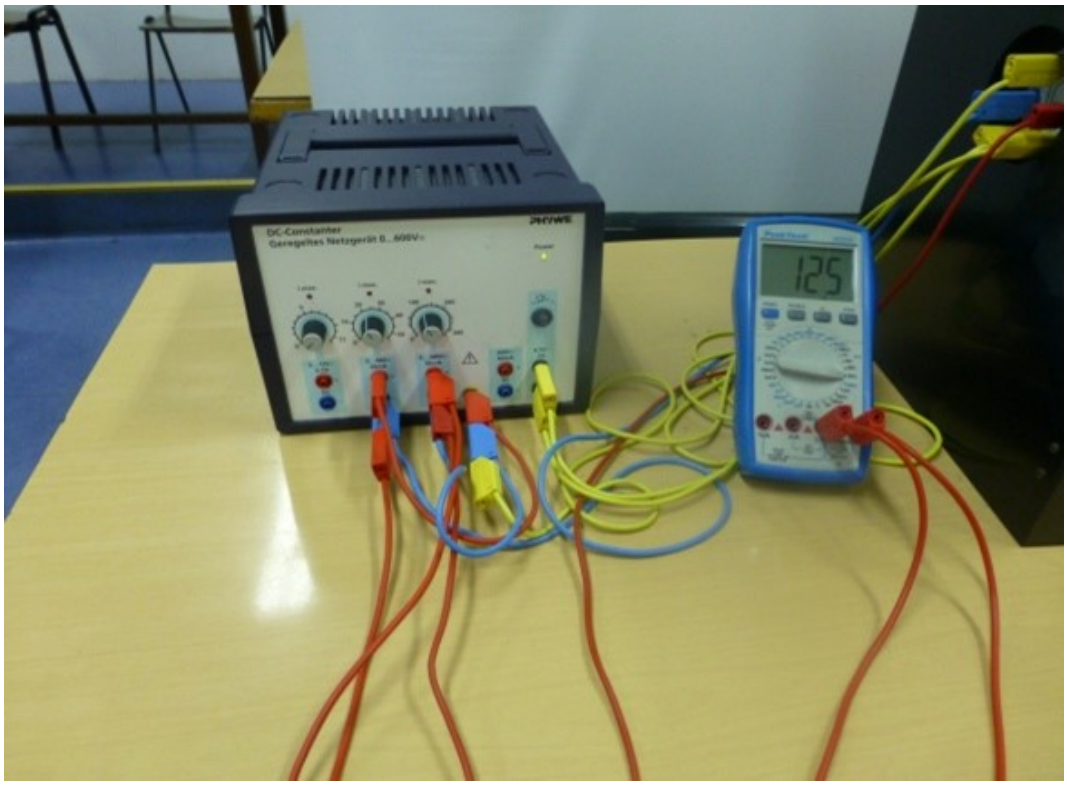


Figure 1.2: Photograph of the experimental setup

they decay to lower energy levels after being excited by the interaction with the accelerated electrons. These interactions are rare (remember that the concentration of neon is low), so that the accelerated electrons do not lose a significant amount of energy. The trails of light left by them allows us to observe the electron trajectories.

The speed at which the electron beam leaves the gun can be changed by changing the accelerating voltage U. In this experiment we are going to use values of U ranging from 120 V to 300 V, with a step of 20 V.

When we run a current through the Helmholtz coil system, a quasi-uniform magnetic field is established in the horizontal direction, in the space between them. If the tube has a correct orientation, the velocity of the electrons is at a right angle with the magnetic field direction, and their motion is circular. The higher the magnetic field induction, the smaller the trajectory radius is.

This particular type of Helmholtz tube is provided with a ladder-shaped measuring device placed along the symmetry axis of the coils. The ladder has four rungs separated by 2 cm distance. The first rung is precisely positioned 4 cm away from the emission point of the electron beam. The rungs are coated with a fluorescent substance so that the impact point can be easily observed.

During this experiment we will set a particular value of the accelerating voltage (and implicitly a value of the speed) and we will change the current through the coils (implicitly the magnetic field) in order to find out what values of I produce circular trajectories of known radius: 2 cm (impact point on the first rung), 3 cm (second rung), 4 cm (third rung) and, finally, 5 cm (fourth rung). As already mentioned, the first radius to be obtained in the increasing order of the current will be Then we will change the accelerating voltage, to have complete experimental data on the dependence of U on I, at each value of the trajectory radius.



Figure 1.3: Photograph of the experimental setup

1.4 Work procedure

Safety precautions:

- **Do not touch the tube and the cables during operation, high voltages are used!**
- **Do not exert mechanical stress on the tube!**
- **Do not increase the current through the coils above 3 A.**

1. If you will be working in a lighted room, place the hood over the e/m apparatus.
2. Before applying the power supply turn down all the control knobs (to the left)!
3. Use the left control knob on the left panel to set a grid voltage of 20 V.
4. Use the right control knob on the left panel to set an accelerating voltage of 120 V. You should be able to observe the vertical trail of the electron beam emitted by the electron gun.
5. Set the maximum value of the current to 3 A by using the right control knob on the right panel.
6. Adjust the current through the coil and observe the effect of increasing the magnetic field on the electron trajectory radius. Write in your data table (an example is given below) the values of I at which the electron beam lands on the 5-th, 4-th, etc. rung of the measuring device. The results will be in the first line of the table.

7. Repeat operations 4 – 6, to complete the table below.

8. Turn down all the control knobs, switch off and unplug.

	$r = 5cm$		$r = 4cm$		$r = 3cm$		$r = 2cm$	
U(V)	$I(\dots)$	$I^2(\dots)$	$I(\dots)$	$I^2(\dots)$	$I(\dots)$	$I^2(\dots)$	$I(\dots)$	$I^2(\dots)$
120								
140								
160								
180							x	x
200							x	x
220							x	x
240							x	x
260							x	x
280							x	x
300							x	x

1.5 Data analysis

The theoretical dependence between the electron accelerating voltage and the intensity is obtained by applying eqs. 1, 3 and 4:

$$U = \frac{32}{125} \cdot \frac{\mu_0^2 n^2}{R^2} \cdot r^2 \cdot \frac{e}{m} \cdot I^2 \tag{1.7}$$

1. Use the data in your table to draw a graph representation of U as a function of I^2. Check that the experimental data you collected agree with the theoretical dependence expressed by eq. 5, for each value of the trajectory radius r. If a data point is too far from it, you may want to repeat that measurement.

2. Use the least square method to find a value of the specific charge for each of the 4 sets of data: $\left(\frac{e}{m}\right)_1$,$\left(\frac{e}{m}\right)_2$,$\left(\frac{e}{m}\right)_3$,$\left(\frac{e}{m}\right)_4$. This means that the estimated value of the specific charge is the arithmetic average of the individual results, $\left(\frac{e}{m}\right)_{av}$.

1.6 Uncertainty calculations

- Calculate the individual uncertainty values for each the four data sets $s_{(e/m)_1}$,$s_{(e/m)_2}$,$s_{(e/m)_3}$,$s_{(e/m)_4}$.
- Apply error propagation to find $s_{(e/m)_{av}}$.

1.7 Conclusions

$\left(\frac{e}{m}\right)_{av}(\dots)$	$s_{(e/m)_{av}}(\dots)$

Compare your results with the result found in CODATA.

1.8 Questions

1. Demonstrate equation 1.7 based on equations 1.3, 1.5, 1.6.
2. Find a value of the proportionality constant between the slope of the dependence $U = f(I^2)$ and $\frac{e}{m}$ at $r = 5$ cm, 4 cm, 3 cm, 2 cm knowing that $R = 0.2m$ and $n = 154$.
3. An electron has a speed of $6.5 \times 10^6 \frac{m}{s}$. Calculate the magnetic field which needs to be applied, so that the trajectory radius is $r = 2cm$.
4. If the magnetic field in question 3 is obtained using the Helmholtz coil in our laboratory, what is the current which needs to be flow through the coils?
5. Can this experiment be performed by simply using an evacuated tube? Justify your answer.
6. What other linear dependence can be used to retrieve the specific charge $\frac{e}{m}$ value from the same sets of data, $(U_i, I_i)_{r_{constant}}$?

Experiment 2

Determination of the detector dead time using the two-source method

2.1 Purpose of the experiment

- Determination of the dead time of a Geiger-Müller counter

2.2 Theoretical context of the experiment

Radioactive decay is one of the most common phenomena occuring in nature. Despite that it was discovered very late because the ionising radiation emitted by radioactive sources cannot be detected directly using our senses. As a result in characterising radioactive sources we have to rely exclusivelly on experimental devices, also called *radioactive detection chains*. A radioactive detection chain has the following components: a detector, a measurement system and auxiliary systems required for their operation.

A simplified description of the way in which radiation detectors give us information about the radioactive decay is the following:

- one or more ionising particles are emitted in each radioactive decay;
- some of these particles will enter a well defined region of the detector, named the *active region* of the detector;
- while moving across (inside) this region, the particle will interact with the material of the detector, creating locally distributed charges;
- these charges are transformed into electrical signals by the measurement system and processed;
- as a result, we obtain information about the particle that interacted with the detector and, indirectly, about the radioactive decay that took place.

The present experiment uses a Geiger-Müller counter. A Geiger-Müller counter is very simple to build and operate. The main component of a Geiger-Müller counter is a metallic tube filled with a noble gas at low pressure. In the center of the tube there is a metallic wire, placed at high positive electric potential. The result of this two-electrode field configuration is a rather simple electric field distribution in the volume of the gas. The field lines are radially distributed, the electric field intensity being much larger in the proximity of the wire (anode) than at the tube (cathode). When a single ionizing particle passes through

the gas volume it will ionize the gas, creating electron-ion pairs. If the electric field is low, the electrons will recombine with the ions thus creating neutral atoms. If the electric field is high, the electron and the ion will start drifting in opposite directions determined by the orientation of the local electric field. While they travel, their energy quickly increases due to the field but they also lose energy by collisions with other electrons/atoms inside the gas, until they eventually reach the electrode. In the proximity of the anode where the electric field intensity is high, these collisions will result in the creation of electron-ion pairs. Each of the the newly created pairs follows the same process as the initial pair, with more and more electrons and ions travelling towards the electrodes. In this cascade process, most of the charged particles collected at the electrodes are not generated directly by the initial ionizing particle. For example, if the ionising particle creates a single electron-ion pair, as soon as the electron gains enough energy it will create (in a collision) another electron-ion pair. As a result, two electrons and two ions will be moving inside the gas in opposite directions. Once any of electrons gains enough energy another electron-ion pair is created. Hence, single electron-ion pair can generate a high number of electron-ions pairs that are moving towards the electrodes. The avalanche process (drived by the electrons) mainly takes place around the center wire because the field intensity is much higher there. Once the electrons are collected, a large spatial distribution of ions (positive charge cloud) appears around the wire (anode). The positive charge cloud creates a shielding effect around the central anode, which lowers the electric field inside the gas. As the electric field gets lower, the generation of more electron-ion pairs slows down, leading to the termination of the avalanche effect. Some time after the first electron-ion pair is created, the gas is again without any free charge and the detector is ready for starting another avalanche.

The charge that moved across the gas region leads to the creation of an electric current pulse that can be read by the external circuit of the detector. The external circuit counts how many such pulses were created, indirectly measuring the number of initial electron-ion pairs and hence the number of ionising particles that entered the detector.

What happens if another ionising particle enters the detector during the time that an avalanche is ongoing? It will generate one or more electron-ion pairs that will start moving as described above. In this case though, this pair will join the other pairs that are making up the avalanche, and, as a result, it will be impossible to say that another particle has arrived. This leads to the idea of *detector dead time*: a minimal time that has to pass after the detection of one particle until another particle can be detected, denoted by τ[1]. For Geiger-Müller counters, the dead time is related to the interval between the moment when the first electron-ion pair was generated and the time when the charge generated by it finished moving through the detector. The dead time depends on the intensity of the electric field (determined by the voltage between the electrodes and the size of the detector) and also on the type of gas inside the detector. The dead time of Geiger-Müller detectors is a few μs.

The dead time influences the total number of detected particles because some of the incident particles will not generate a signal. Consider a source that produces N particles interacting with the detector in a time interval T. Out of these particles, only N' will generate a signal (the particles that arrive during the time an avalanche is already ongoing are not detected). If the number of particles which arrive at the detector does not vary in time (which is usually true for measurements performed using radioactive sources), the number of lost particles in time T is proportional to the time the dector was "dead" and

[1]there is also a contribution from the associated electronics, which we will ignore for this experiment

with the rate at which the particles are arriving at the detector:

$$N - N' = (N'\tau) \cdot \frac{N}{T}$$

We can rewrite this relation by introducing $n = \frac{N}{T}$ to denote the *real count rate* and $n' = \frac{N'}{T}$ to denote the *experimental count rate*:

$$n - n' = n \cdot (n'\tau)$$

and from that we can see that:

$$n = \frac{n'}{1 - n'\tau} \tag{2.1}$$

Equation 2.1 can be used to calculate the real count rate when the dead time of the detector is known. On the other hand, the same equation can be used to calculate the deadtime of the detector using two radiation sources by performing four measurements: one measurement with each of the sources, a measurement with both sources and a measurement without any of the sources. The measurement without sources is necessary because we need to remove from the analysis the contribution of the counting rate due to the background radiation. The background radiation affects equally all measurements.

We use the following notations:

- n_{1b} the real count rate when source number 1 is used;
- n_{2b} the real count rate when source number 2 is used;
- n_{12b} the real count rate when both sources 1 and 2 are used;
- n_b the real count rate when no source is used (and we are only measuring the radiation background).

For each of these measurements, the real count rate is related to the experimental count rate through an equation similar to 2.1.

Because of the way we designed the experiment, a simple relation exists between the four real count rates mentioned above:

$$n_{12b} + n_b = n_{1b} + n_{2b} \tag{2.2}$$

This relation can be rewritten using equation 2.1, leading to a relation between the experimental count rates and the deadtime, which can be solved for τ:

$$\tau = \frac{n'_{1b} + n'_{2b} - n'_{12b} - n'_b}{2\left(n'_{1b} \cdot n'_{2b} - n'_{12b} n'_b\right)} \tag{2.3}$$

This equation expresses the deadtime τ only in terms of the measurable (experimental) count rates (n'_{1b}, n'_{2b}, n'_{12b}, n'_b), hence providing a way of determining its value.

2.3 Description of the experiment

The experiment uses the following items:

- Geiger-Müller counter;

- auxiliary system for the Geiger-Müller counter;
- two radioactive sources, placed in a closed enclosure (you need to ask the assisting professor to bring you the source); the lid of the enclosure is sliding so that one, the other, or both sources can be uncovered;
- counting system: counts the number of pulses from the detector.

2.4 Work procedure

1. Measure the background count rate.
2. Ask for the sources. (The assistant professor will place the sources in the proper position and explain how the sources should be manipulated.)
3. Uncover one of the sources and measure the count rate; use the front panel setting to choose the duration of the measurement of 60 or 100 seconds.
4. Uncover the other source (only one source should be uncovered) and measure the count rate.
5. Uncover both sources and measure the count rate.
6. Cover all sources and measure the count rate.
7. Repeat steps 3-6 until you have at least three sets of measurements.

Example of table:

No	T(...)	N'_b (...)	n'_b(...)	N'_{1b}(...)	n'_{1b}(...)	N'_{2b}(...)	n'_{2b}(...)	N'_{12b}(...)	n'_{12b}(...)	$\tau \pm \sigma_\tau$(...)
1										
2										
⋮										

2.5 Data analysis

Calculate the dead time for each of the measurement sets.

2.6 Uncertainty calculations

Calculate the uncertainty of the individually measured dead-time using the uncertainty propagation formula applied to relation 2.3 for only one of the measurement sets. (Answering questions 3, 4, 5 and 6 will provide useful hints on how to simplify the calculations.)

2.7 Conclusions

Measured dead-time(τ)(...)	Uncertainty (...)

Note: report only the value from the individual measurement for which you have also calculated the uncertainty. Explain why averaging the values obtained in the different measurements is not correct.

2.8 Questions

1. Explain formula 2.2.
2. Write the uncertainty propagation formula for a function of two variables;
3. Using the uncertainty propagation formula calculate:
   (a) the uncertainty σ_f, if $f(x, y) = x + y$;
   (b) the relative uncertainty ϵ_f, if $f(x, y) = x \cdot y$;
   (c) the relative uncertainty ϵ_f, if $f(x, y) = x/y$;

   Use the notations:

   - σ_x - uncertainty of x;
   - ϵ_x - relative uncertainty of x: $\epsilon_x = \sigma_x/x$.
4. Write the uncertainty propagation formula applied to equation 2.3. Can you simplify it (and/or the calculations) using the results of question 3?
5. How many significant digits should the uncertainty be written with (see also the the note at the end of this section)? How many significant digits should the final result be written with?
6. What is the uncertainty of a counting experiment (assuming a Poisson distribution) when the number of measured counts is N? What is the uncertainty of the count rate? (see Annex A.3)
7. (Optional) Is it possible for the measured deadtime to be negative? Justify your answer using equation 2.3.

Note: The official document describing the treatment of measurement data and their uncertainties is "Evaluation of measurement data - Guide to the expression of uncertainty in measurement" published by BIPM, JCGM 100:2008. (you can find this document by searching for "GUM BIPM"). Section 7.2.6 (reproduced bellow) defines the number of digits to be used in expressing the uncertainty in measurements. **For the purpose of our measurements, we can use a single significant digit (or two if that significant digit is 1)**

"7.2.6 The numerical values of the estimate y and its standard uncertainty $u_c(y)$ or expanded uncertainty U should not be given with an excessive number of digits. It usually suffices to quote $u_c(y)$ and U [as well as the standard uncertainties $u(x_i)$ of the input estimates x_i] to at most two significant digits, although in some cases it may be necessary to retain additional digits to avoid round-off errors in subsequent calculations.

In reporting final results, it may sometimes be appropriate to round uncertainties up rather than to the nearest digit. For example, $u_c(y) = 10,47$ mΩ might be rounded up to 11 mΩ. However, common sense should prevail and a value such as $u(x_i) = 28,05$ kHz should be rounded down to 28 kHz. Output and input estimates should be rounded to be consistent with their uncertainties; for example, if y = 10,05762 Ω with $u_c(y) = 27m\Omega$, y should be rounded to 10,058 Ω. Correlation coefficients should be given with three-digit accuracy if their absolute values are near unity. "



Figure 2.1: Photo of the auxiliary system for the Geiger-Müller counter. For clarity, the names of the buttons used in the measurement are added in red to this image.

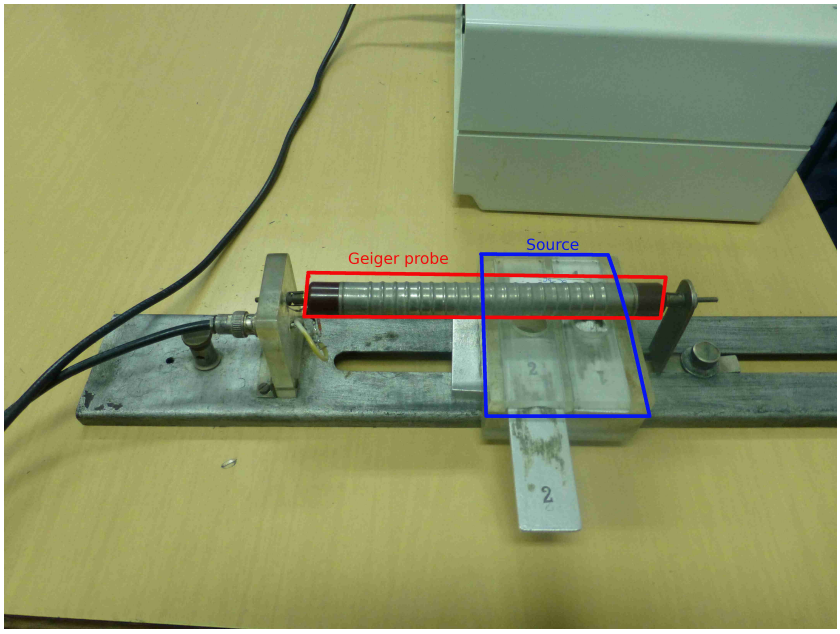


Figure 2.2: Photo of the Geiger-Müller probe, together with the source holder.

Experiment 3

Determination of Rydberg constant

3.1 Purpose of the experiment

- Determination of the wavelengths in the Balmer series of atomic hydrogen by comparison with mercury spectrum;
- Using spectral data to calculate Rydberg constant.

3.2 Theoretical context of the experiment

Light emitted by atoms can be studied using discharge lamps. A discharge lamp is an electronic device in which a collection of atoms at low pressure are excited by applying a high voltage. Some of the electrons will pass into high energy states and, after a short time (the excited state lifetime), they fall back on lower energy states. The lowest energy state in an atom is called the ground state and all the others are excited states. Any transition to a lower energy level, whether it is the ground state or a low excited state, is accompanied by light emission. The emitted frequency $f_{n_i n_f}$ is proportional to the difference between the involved energy levels:

$$f_{n_i n_f} = \frac{E_{n_i} - E_{n_f}}{h}, \tag{3.1}$$

where E_{n_i}, E_{n_f} are the energy values of the upper (initial) and lower (final) levels and and $h = 6.626 \times 10^{-34}\ Js$ is Planck's constant.

Light emitted by any discharge lamp will contain many frequencies. They represent the emission spectrum of the excited atom. An atom with more electrons will have more energy levels and thus a more complex emission spectrum. In fact, the emission spectrum of an atom is the "fingerprint" which allows us to detect its presence in an electric discharge, even in small quantities. The simplest spectra are obtained from hydrogen and hydrogen-like atoms because they have a single electron. Bohr's planetary model can be applied to obtain an expression of the frequencies which exist in their emission spectrum and, correspondingly, the wavelengths, since , since $\lambda = c/f$ (c is the speed of light in vacuum. According to this model, the energy E_n of an electron on a certain energy level n is given by:

$$E_n = -\frac{1}{8}\frac{m_e e^4}{\epsilon_0^2 h^3} \times \frac{1}{n^2},\ n = 1, 2, 3, \ldots \tag{3.2}$$

where $\epsilon_0 = 8.854 \times 10^{-12} F/m$ is the electric field constant in vacuum, $e = 1.6 \times 10^{-19}\ C$ - the elementary charge, and $m_e = 9.1 \times 10^{-31}\ kg$ the rest mass of an electron. Any bound

state of an atomic electron has a negative value. The lowest energy state in the hydrogen atom is

$$E_n = -\frac{1}{8}\frac{m_e e^4}{\epsilon_0^2 h^3} = -13.6\ eV, \tag{3.3}$$

in which a conversion from joules to electronvolts was applied ($1\ eV = 1.6 \times 10^{-19}\ J$). The frequencies in the emission spectrum of a hydrogen atom are expressed as:

$$f_{n_i n_f} = \frac{E_{n_i} - E_{n_f}}{h} = \frac{1}{8}\frac{m_e e^4}{\epsilon_0^2 h^4} \times \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), n_f < n_i \tag{3.4}$$

Before Bohr produced his atomic model, spectroscopists had already noticed that the emission lines of a particular atom can be grouped in spectral series which fit a certain pattern. For example, by examining the visible lines of hydrogen, Balmer retrieved in 1885 a simple formula to describe the characteristic pattern of this series. In 1888 Rydberg gave a more general formula, which could be used to calculate all the emission lines in the hydrogen spectrum:

$$\frac{1}{\lambda_{n_i n_f}} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), n_f < n_i \tag{3.5}$$

Rydberg's result was expressing the inverse wavelength of a particular hydrogen line in terms of an empirical constant R_H (later called Rydberg constant) and two integer numbers, n and m , identifying the series and the line. Balmer formula proves to be a particular case of Rydberg formula, for $n = 2$. At that point in time it was not known *why* this formula worked, since it was not a theoretical result.

In 1913 Bohr introduced the concept of discrete energy levels in an atom. His planetary model found an excellent confirmation in Balmer and Rydberg work. Indeed, we can easily write Bohr model result (3) in a form similar to Rydberg formula:

$$\frac{1}{\lambda_{n_i n_f}} = \frac{E_{n_i} - E_{n_f}}{c \cdot h} = \frac{1}{8}\frac{m_e e^4}{c\epsilon_0^2 h^4} \times \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), n_f < n_i \tag{3.6}$$

According to eq 5, Rydberg constant has a theoretical expression:

$$R_H = \frac{1}{8}\frac{m_e e^4}{c\epsilon_0^2 h^4}, \tag{3.7}$$

which is in excellent agreement with the value obtained by fitting experimental spectral data to Rydberg formula. Also, the integers n_i and n_f get a clear physical significance: : they represent the principal quantum numbers of the final and the initial energy states of the transition that produces light of wavelength $\lambda_{n_i n_f}$. Keeping in mind the physical interpretations given above, Balmer formula reads:

$$\frac{1}{\lambda_{n_i,2}} = R_H \left(\frac{1}{2^2} - \frac{1}{n_i^2}\right), n_i = 3, 4, \ldots \tag{3.8}$$

It predicts the wavelengths in the visible region of the hydrogen spectrum that results from transitions to the $n_f = 2$ final excited state. There are several other spectral series in the hydrogen atom as it were: Lyman (transitions to $n_f = 1$, in ultraviolet), Paschen ($n_f = 3$), Bracket ($n_f = 4$), Pfund ($n_f = 5$) . The last three series contain lines in the infrared regions of the electromagnetic spectrum.

3.3 Description of the experiment

- In this experiment we are going to determine a value of Rydberg constant by fitting Balmer spectral series formula (3.8) to the wavelengths λ_{H_α}, λ_{H_β}, λ_{H_γ}, λ_{H_δ}, λ_{H_ϵ} of the lines in the Balmer spectral series of atomic hydrogen.

- In order to find these wavelengths, we apply a spectroscopical technique. It consists in comparing the unknown spectrum (hydrogen) to a known spectrum (mercury). The comparison is correct only if the two spectra are recorded in the same experimental set up (in this case - a prism spectrograph), by placing in turn the mercury and hydrogen light sources at exactly the same position. Two different images are recorded in parallel (for example, by translating vertically the photographic plate). In fact, a similar experiment, called "Study of light dispersion - the spectroscope" is available in the laboratory. However, the optical spectra are viewed instead of being recorded. In this experiment we use a spectrogram in which the emission lines are dark on a white background, as seen in Figure 3.1.

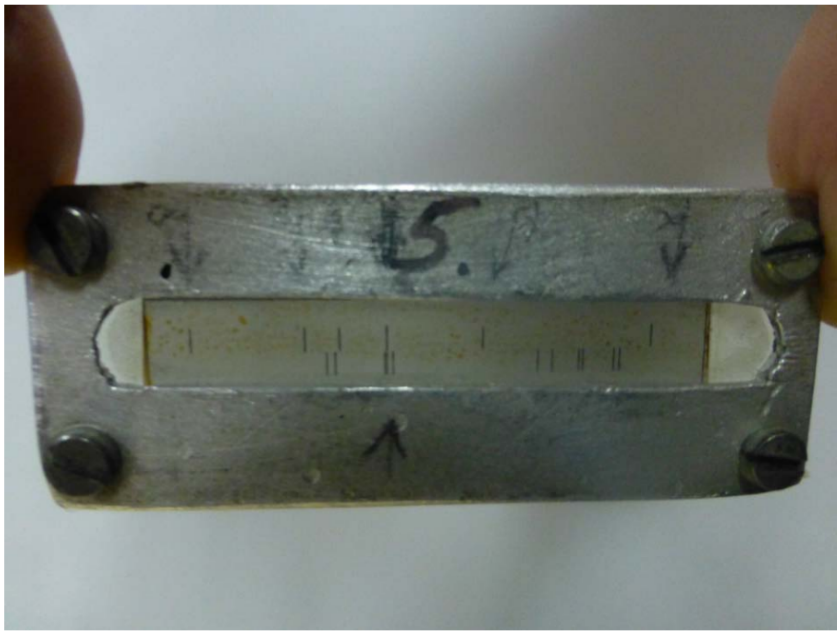


Figure 3.1: Photo of the spectrogram used in the experiment.

Since the two spectra were obtained by passing light through the same prism, according to Cauchy's dispersion formula, the position of the individual lines $x(\lambda)$ is given, in the first approximation, by the equation

$$x(\lambda) = A + \frac{B}{\lambda^2}. \tag{3.9}$$

The experiment uses the following items:

- spectrogram containing mercury and hydrogen spectra
- optical microscope
- XY translation mechanical stage equipped with

a. device which keeps the spectrogram in place

b. graduated locator (main scale divided in mm and a vernier allowing a 0.1 mm location precision).



Figure 3.2: Experimental setup: microscope with holder and mounted spectrogram.

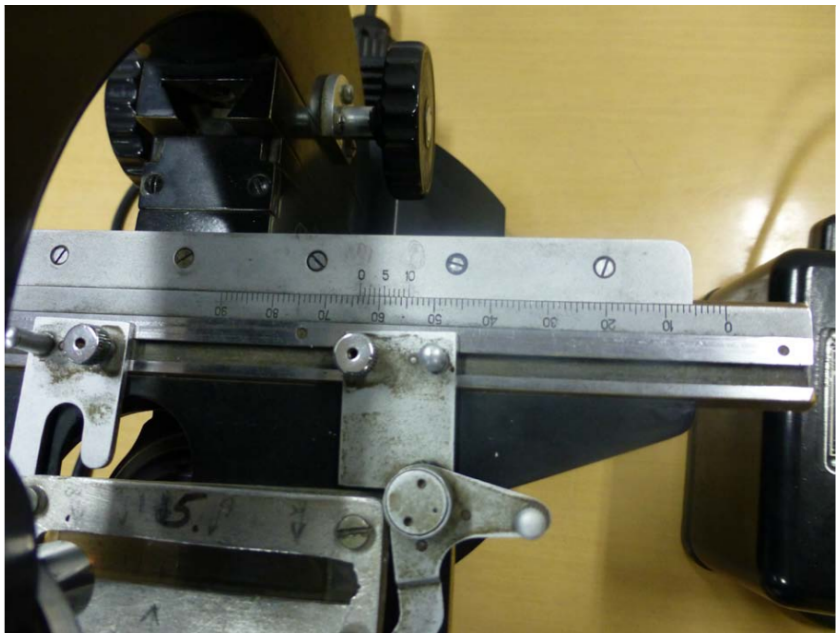


Figure 3.3: Photo of the graduated locator.

3.4 Work procedure

1. Mount the spectrogram device on the stage and focus the microscope to get a clear image of the mercury and hydrogen spectra. Make sure that the spectral lines are

parallel to the reticle in the microscope eyepiece.

2. Identify which of the two spectra belongs to mercury and bring it into your field of view by using the translational control knobs in the x (perpendicular to the lines) and y (parallel to the spectral lines) directions.

3. Move the stage until each mercury line is under the reticle and record its position on the locator (precision 0.1 mm); fill the data in the correct column of Table 1 (this implies a correct association of the known wavelength to the lines that you are measuring. Hint: look up the group of three lines which are close together (433.9 nm, 434.7 nm, 435.8 nm). On one side of this group there are only two lines, on the other side there are six lines. Start with the smallest wavelength and make sure that the associated reading of the locator is also increasing with the wavelength. If it is not, just flip the spectrogram plate.

4. Repeat the procedure to record all the hydrogen spectral lines (Table 2). In order to identify correctly the lines you may use the fact that the γ line of hydrogen lines up with the 434.7 nm line in the mercury spectrum (see figure 3.2). You will also need to use the y translational knob to get the hydrogen spectrum in your field of view.

An example of data tables is given bellow:

$\lambda_{Hg}(nm)$	404.7	407.8	433.9	434.7	435.8	535.4	546.1	577.0	579.0	612.3	623.4
$x_{Hg}(mm)$											

Table 3.1:

Spectral line	H_α	H_β	H_γ	H_δ	H_ϵ
Principal number (n_i)					
$x_H(\ldots)$					
$\lambda_H(\ldots)$					

Table 3.2:

3.5 Data analysis

1. Use data in Table 3.1 to draw a graph representation of x_{Hg} as a function of λ_{Hg}^{-2}. According to eq. 3.9 this is a straight line of slope B and intercept A. Check that your experimental data are in agreement with this. If a data point is too far, repeat the measurement of that particular line.

2. Apply least square method to retrieve A and B, then use eq. 3.9 to find the walelengths λ_H of the Balmer spectral lines of hydrogen and fill the last line of Table 3.2.

3. Draw a graph representation of λ_H^{-1} as a function of n_i^{-2} . According to eq. 3.8, this is a straight line whose slope is equal to Rydberg constant. Use least square method to find the Rydberg constant.

3.6 Uncertainty calculations

Calculate the uncertainty of the attenuation coefficient using the uncertainty of the slope calculated from the least square method.

3.7 Conclusions

Rydberg constant $(R_H)(\dots)$	Uncertainty $(\dots)$

3.8 Questions

1. What is the relation between the energy and the frequency of a photon? What about the dependence of energy on wavelength?

2. Can one see radiation in the Lyman series? Justify your answer. Paschen spectral series.

3. Calculate five wavelengths in the visible light range for the hydrogen gas emission by applying eq. 3.8. Record your results in the increasing of order in the first 3 columns of the data table bellow. Use $R_H = 1.09737 \times 10^7 m^{-1}$

n_i	n_f	$\lambda_H(nm)$	Colour
$\dots$	2		
$\dots$	2		
$\dots$	2		
$\dots$	2		
$\dots$	2		

4. Find the colour of the lines in the Balmer series and fill the 4-th column in the table above. x_{Hg} is on the abscissa.

Experiment 4

Study of electron diffraction by polycrystalline graphite

4.1 Purpose of the experiment

- Observing the phenomenon of electron diffraction by a graphite polycrystal
- Determining values of the crystalline lattice plane spacing for graphite

4.2 Theoretical context of the experiment

Diffraction is a typical light wave phenomenon. When a light wavefront is incident on small obstacles/apertures, it is "broken into pieces" (diffringere in Latin) and the "fragments" are scattered behind the diffracting object. The result is a redistribution of the incident light energy into zones of bright intensity (maxima) and dark intensity (minima), named fringes, forming a diffraction pattern. This is composed of alternate fringes whose geometry (point-like, rectangular, etc.) depends on the geometrical details of the diffracting object. For diffraction to occur, the linear dimension of the obstacle/aperture needs to be very small, of up to roughly ten times the incident wavelength.

In a similar way, electron diffraction can be described as a phenomenon in which a beam of electrons produces a diffraction pattern when it is incident on an adequate obstacle/aperture system. This means that, under certain conditions, electrons (which are known to be particles of mass $= 9.1 \times 10^{-31} kg$) are diffracted just as light waves. The first to predict such a wave-like behavior of material particles was Louis de Broglie (1924) who produced the hypothesis that all particles in motion have an associate wave whose wavelength is

$$\lambda = h/p \tag{4.1}$$

where $h = 6.63 \times 10^{-34} J \cdot s$ is Planck's constant and p is the momentum of the particle of mass m and speed v.

Given the smallness of h, even a tiny macroscopic body will have an associate wavelength which is much less than the dimension of any physically achievable obstacle/ aperture, no matter how fast it is! (Note that the speed itself cannot be increased indefinitely: according to the second postulate of the theory of relativity, the speed of any object is limited to the speed of light in vacuum, which is $3 \times 10^8 m/s$). "Beams" of macroscopic bodies are not expected to be diffracted because there are no obstacles/apertures small enough to produce this effect.

However, electrons have a very small mass, so they can have an associate wavelength of the order of magnitude of a extremely small obstacle, such as an atom or molecule. An extremely-small-size-aperture example is the separating distance between the crystal planes in a crystalline material. It is reasonable to consider that electrons will undergo diffraction, should de Broglie hypothesis be valid.

Between 1921 and 1927 the American physicists C. Davisson and L. Germer were conducting experiments of electron beam scattering by crystalline nickel to study its surface properties. As a by-product of their experiments they observed that reflected electrons produce a maximum of diffraction corresponding to the associated de Broglie wavelength. This was the first experimental evidence of the fact that electrons exhibit wave-like behavior. Subsequent measurements and calculations validated quantitatively de Broglie hypothesis.

Let us consider a beam of electrons obtained by acceleration under a small potential difference U. According to the kinetic energy theorem, we get

$$eU = \frac{mv^2}{2} = \frac{p^2}{2m} \tag{4.2}$$

where e is the magnitude of the electron charge ($1.6 \times 10^{-19} C$), v and p are the electron speed and momentum, respectively, and m is its mass ($9.1 \times 10^{-31} kg$). De Broglie hypothesis states that the wavelength associated to this monoenergetic particle beam is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meU}}. \tag{4.3}$$

When these waves are incident on a crystalline lattice, they are reflected by its parallel planes containing the atoms. For simplicity, in 4.2, two such planes separated by a distance d are represented. The reflected waves will interfere constructively (add up) if their path difference ($2\Delta = 2d \sin\theta$) is an integer number of wavelengths:

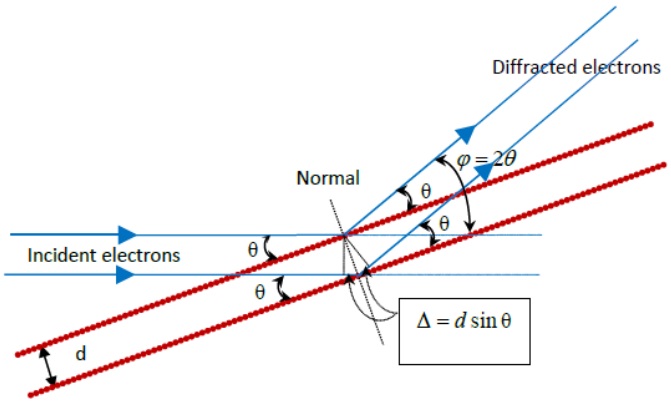


Figure 4.1:

$$2d \sin\theta = n\lambda, , \tag{4.4}$$

where $n = 1, 2, ..$ is the order of diffraction. Equation 4.4 represents the condition for a maximum of diffraction. From 4.2, one can also observe that the diffracted beam is deflected under an angle φ which is twice the angle of incidence, $\varphi = 2\theta$. Unless condition 4.4(is fulfilled, the reflected waves do not interfere constructively. This means that for a certain distance d and a certain angle of incidence θ, a quasi-monoenergetic electron beam will

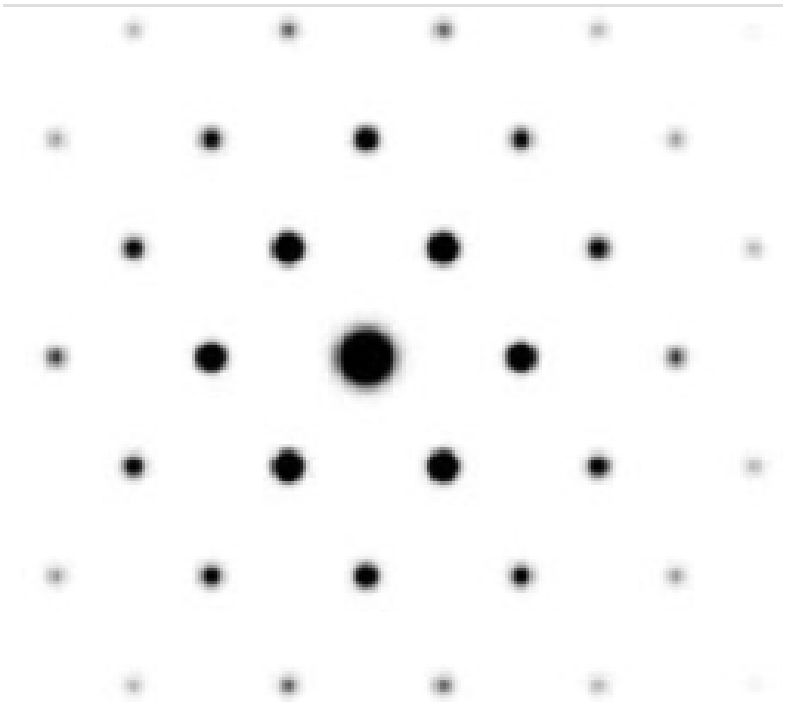


Figure 4.2: Electron diffraction pattern on a single crystal - graphene

produce a maximum spot (localized region in the detecting area, where electrons will be recorded).

However, in a crystal there is always a finite number of different possible lattice plane spacings, particular to the type and morphology of the crystal. Also, crystal planes have different spatial orientations. Each different value of the spacing d will give rise to a spot (maximum of interference) in a different direction. For example, if another value of d is smaller than the one represented in 4.2, the angle of deflection φ will be larger, so that the product $2d\sin\theta$ can stay the same for the same associated wavelength λ of the incident electron. Consequently, when a quasi-monoenergetic beam of electrons passes through a single crystal, it will produce a diffraction pattern of spots, as shown, for example, in 4.2. Graphene is a single crystal with hexagonal symmetry. Measuring such a diffraction pattern can be difficult because you need to rotate either the detector or the crystal in order get the correct incidence at which the condition for a maximum (eq. 4) is fulfilled.

A solution to this problem is provided by using polycrystals instead of single crystals. Polycrystals are solid object composed of randomly oriented crystalline regions, called crystallites. The crystallites are small single crystalline regions. It is like exposing simultaneously the incident electrons to all the possible spatial orientations of the family of crystalline planes separated by a certain distance d_1, for example. All the spots produced by them are going to be located on a circle because they correspond to the same value of the diffraction angle, θ. Another family of planes, separated by another characteristic d_2, having all possible orientations, will produce another diffraction ring, etc. Hence, the diffraction pattern of electrons by a polycrystalline substance will produce a ring pattern which carries information on the single crystal structure.

In our experiment we are using a polycrystalline graphite target. Single crystalline graphite is composed of stacked layers of carbon atoms in a hexagonal pattern, as presented in Figure 3a. The entire crystal can be obtained by "cloning" indefinitely a unit cell in independent directions (see, for example, Figure 3b). The bond between neighboring

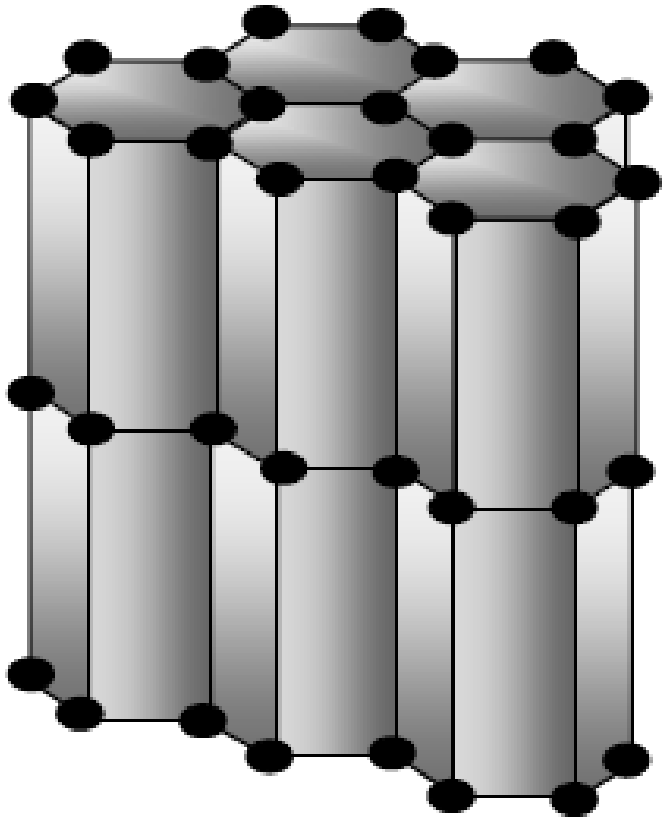


Figure 4.3:

layers is weak, and the layers can easily slide against each other. In figure 3c you can see a planar projection of a graphite crystal. A family of planes separated by distance 1d is represented by black lines. However, in that crystal the incident electrons can also "see" other families of planes,defined by distance d_2 (green line) or d_3 (red lines). For simplicity, in the last two cases, only a pair of planes was drawn, but families of planes can be easily obtained by translation, as seen in the d_1 case. Note that $d_1 > d_2 > d_3$, which means that the values of the diffraction angles will be $\theta_1 < \theta_2 < \theta_3$ (as explained before), thus producing diffraction rings of diameters $D_1 < D_2 < D_3$. In fact, in our experimental setup the third diffraction ring cannot be observed, because its diameter D_3 exceeds the diameter of the detecting area.

4.3 Description of the experiment

In this experiment we are going to:

- observe the diffraction pattern produced by a beam of accelerated electrons incident on a polycrystalline graphite target.
- measure the diameters 1D and 2D of the first order diffraction rings (maxima) which can be seen with our experimental setup at several values of the electron accelerating voltage, U. 1D and 2D correspond to the first two lattice plane spacing values 1d and 2d, which will eventually be retrieved by data processing.

The experimental set up is composed of an electron diffraction tube and a high voltage power supply (10 kV). The electron diffraction tube hosts its components in an evacuated (very low pressure gas) clear glass bulb (see Figures 4 and 5). The components are:

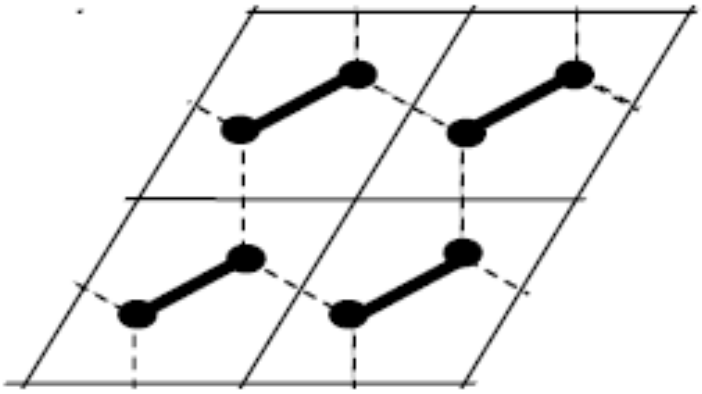


Figure 4.4:

(a) the electron gun - designed to produce a collimated electron beam. It is composed of:

- filament (heated when a voltage is applied between the F1, F2 sockets and a current runs through it; the heat necessary for the cathode to release electrons is provided by the filament);
- cathode (metallic oxide surface which emits electrons by thermionic effect when heated);
- cathode (C) / anode (A) system which is polarized by applying an accelerating voltage U; the electrons are accelerated in the electric field established between C and A;
- supplementary electrodes placed between C and A which are designed to produce a parallel electron beam at point A.

(b) the target - a graphite polycrystalline foil placed close to the anode (practically at point A).

(c) the observing screen - a fluorescent layer deposited on the inner side of the glass bulb (the white region of the bulb). What we actually see is the visible fluorescence radiation produced by the interaction of electrons with the screen.

We will denote by L the distance from the target to the screen in the direction of the electron incidence on the target. $\varphi = 2\theta$ is the half opening angle of the cone defined by the direction of the diffracted electrons (see the theoretical considerations, Figures 1 and 4). D is the measured diameter of any of the two observed rings. Then, the following equations hold with a good approximation:

$$tan(2\theta) = \frac{D}{2L} \approx \sin(2\theta) \approx 2\sin\theta \tag{4.5}$$

For the first order of diffraction 1n= the condition for a maximum (4.5) reads:

$$2d\sin\theta = \lambda, \tag{4.6}$$

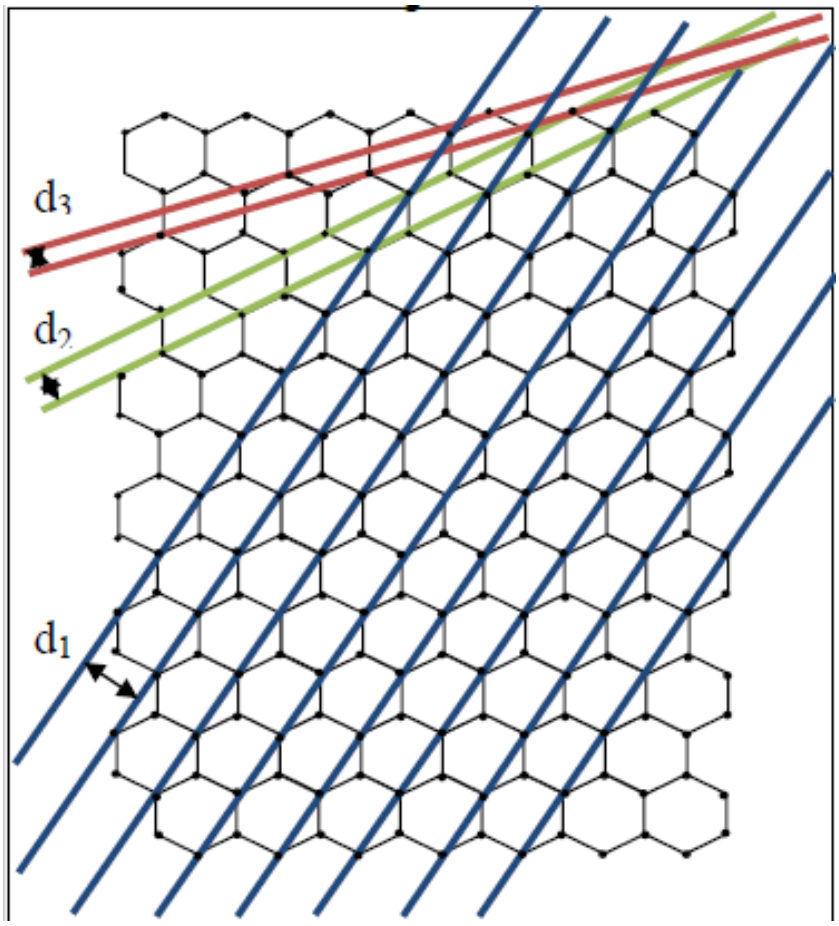


Figure 4.5:

where the associated electron wavelength is given by eq. 4.3. By combining eq. 4.3, 4.5 and 4.6 we get a relation between the measured diameter of the diffraction ring and the accelerating voltage needed to obtain it:

$$D = \frac{2Lh}{d\sqrt{2me}} U^{-1/2}. \tag{4.7}$$

In this experiment $L = 13.5cm$. By plugging the constant values into eq. 4.7, we get that

$$D = \frac{3.3 \times 10^{-10}}{d} U^{-1/2} SIunits \tag{4.8}$$

4.4 Work procedure

Safety precautions

- **Do not operate the tube at a voltage higher than 5 kV - X radiation will be produced by the accelerated electrons!**
- **Do not touch the tube and the cables during operation, high voltages are used!**
- **Do not exert mechanical stress on the tube - there is danger of implosion!**

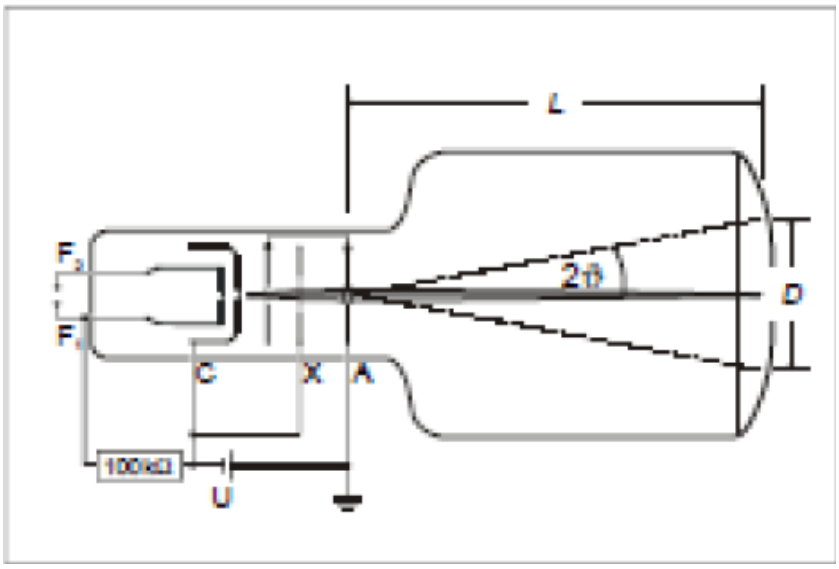


Figure 4.6:

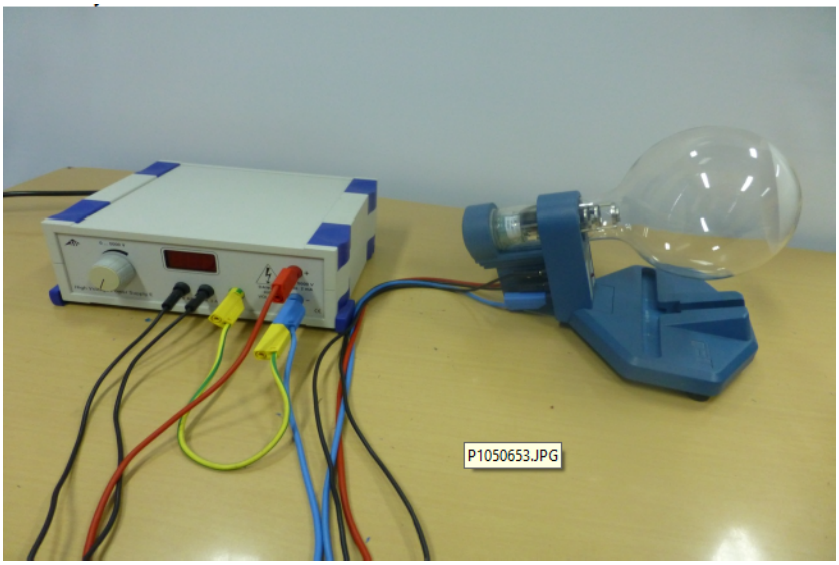


Figure 4.7:

- **The bright spot in the center of the screen can damage the fluorescent layer. Reduce the accelerating voltage as soon as possible after each reading!**

1. Make sure the power supply is off and turn the accelerating voltage control knob all the way down (to the left).

2. Turn the power switch on and wait a minute till the cathode surface reaches thermal stability.

3. Slowly increase the high voltage U until the ring pattern appears on the screen, then you will continue to apply values of the accelerating voltage up to 5 kV. PLEASE DO NOT EXCEED THIS VALUE, because at higher electron energy X rays are produced!

4. Choose a set values of the accelerating voltage U and for each chosen value, measure the corresponding inner and outer diameters of the two diffraction. Start with any value which allows a correct measurement.

5. Write data in a table. An example is given below (but using this table head is not mandatory). For a good precision of the final results we recommend at least 5 values of the accelerating voltage. **Note:** You can experiment with the accelerating voltage to get the best data points. However, you need to remember not to expose the target and the screen to fast electrons for a long time. Reduce U after each measurement.

6. Turn down the accelerating voltage and unplug.

$U(\dots)$	$U^{-1/2}(\dots)$	$D_1(\dots)$			$D_2(\dots)$		
		Inner	Outer		Inner	Outer	
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
U_{max}							

4.5 Data analysis

- Use the data in your table to draw graph representations of $D_{1,2}$ as a function of $U^{-1/2}$. Check that the experimental data you collected agree with the theoretical dependence expressed by eq. 4.8. If a data point is too far from it, repeat that particular measurement.
- Use the least square method to find the lattice spacing values d_1 and d_2.

4.6 Uncertainty calculations

Use the uncertainty propagation formula to calculate the uncertainty of the lattice spacing values d_1 and d_2 from the uncertainty of the slope obtained by the least square method.

4.7 Conclusions

$d_1(\dots)$	Uncertainty of $d_1(\dots)$	$d_2(\dots)$	Uncertainty of $d_2(\dots)$

Compare your results with $d_1 = 2.13 \times 10^{-10} m$ and $d_2 = 1.23 \times 10^{-10} m$.

4.8 Questions

1. Consider a grain of sand ($m = 3 \times 10^{-8} g$)moving at the speed of sound in air ($v = 340 m/s$). Calculate the de Broglie wavelength and compare it to the carbon atomic diameter ($\approx 2.2 \times 10^{-10} m$) and to the largest plane spacing in a carbon hexagonal lattice ($d_1 = 2.13 \times 10^{-10} m$). Is it possible to obtain a diffraction pattern by shooting grains of sand at a carbon crystal lattice?

2. Repeat the calculation in Question 1 by using an electron accelerated under a potential difference of 2 kV as a replacement for the grain of sand. Is it possible to obtain a diffraction pattern by shooting electrons at a carbon crystal lattice?

3. Calculate the speed and the associated wavelength of an electron accelerated under a p.d. of 3 to 5 kV with a step of 0.5 kV and write the results in a table. Was it correct to use the nonrelativistc formula for the kinetic energy in eq. 2?

4. Explain how does this experiment confirm de Broglie hypothesys?

5. Consider a maximum measured value of 6 cm for the diameter of the second diffraction ring. Is the approximation used in $\tan(2\theta) \approx 2\sin\theta$ (eq. 5) still valid? Justify your answer.

6. What type of a dependence exists between the measured diameter value D and the inverse square root value of the accelerating voltage $U^{-1/2}$, $D = f(U^{-1/2})$?

7. Figure out a linear dependence between the accelerating voltage U and a power of D, $U = f(D^x)$. In such a linear representation what is the equation which connects the slope to the unknown quantity d ?

Experiment 5

Determination of the source activity using the solid angle method and a scintillation detector

5.1 Purpose of the experiment

- Calculating the activity of a radioactive source.

5.2 Theoretical context of the experiment

Activity of a radioactive source tells us the rate (number of decay events that occur every second) at which the isotope decays.

When a nuclear decay takes place, the nucleus is transformed into another nucleus which is more stable (has lower total energy), the extra energy being emitted as radiation. This experiment uses a radioactive source containing ^{137}Cs atoms. Every time one of these nuclei decays it emits one photon, in a random direction. Placing a device sensitive to the incoming photons (a radiation detector) in the vecinity of the source we can detect these photons and measure the source activity. The detector will convert the energy deposited inside its sensitive volume into an electric pulse, which is then counted by an electronic system.

In general, the number of counts in a certain time interval depends on a few factors:

- **the source type:** each radioactive isotope has a characteristic decay scheme. Some isotopes will emit one photon(*e.g.* ^{137}Cs), others will emit two photons (*e.g.* ^{60}Co) or even more(*e.g.* ^{152}Eu);
- **the source activity:** measures the rate at which the decay events occur;
- **detection efficiency:** measures the probability that a photon passing through the sensitive volume of the detector will interact with the detector and produce a measurable signal. The detection efficiency depends on the type of incoming particles (photons, electrons, neutrons, etc.) and their energy;
- **the distance between the source and the detector.**

These dependences can be written as:

$$n = \frac{N}{\Delta t} = \Lambda \cdot \frac{\Omega}{4\pi} \cdot \epsilon \cdot S \tag{5.1}$$

where:

- N is the number of counts in time Δt;
- n is the count rate of the detector;
- Λ is the source activity;
- Ω is the solid angle;
- ϵ is the detection efficiency;
- S is a number which shows how many photons are emitted during each decay event.

Equation 5.1 establishes a direct relation between a measurable quantity (the number of counts) and the value we are interested in calculating from the experiment (the activity of the source). In this equation, all the factors except the solid angle are predermined because they are either natural constants or they are related to the details of the detector construction. The values of these factors are obtained in other experiments and they will be considered as "known" for the present experiment.

5.2.1 Solid angle

We know that the same object will occupy more of our field of view when the distance between the object and the eye is decreasing. This is not due to the object getting larger or of a modification of our eye, but it is a geometrical effect. In a two dimensional world, we would say that the angle between two lines starting from our eye and going through the geometrical extremes of the object is increasing as the object is coming closer. But we live in a three-dimensional geometrical world, so we need a way to describe the same effect for solid objects of any dimensions. The adopted solution is to express the effect using the **solid angle**, which is a measure of how much of our field of view is "covered" by the object.

In the International System of Units (SI), the solid angle is expressed in a dimensionless unit called steradian, which has the symbol sr. The solid angle is defined in relation to a point, called the *apex*. Its value is related to the ratio between the covered area on a sphere and the square of the distance to the apex.

Imagine that you (your eyes) are sitting at the center of a one meter radius sphere and you are free to look around in any direction. An object is brought in your field of view. It will "block" you from seeing a certain area of the sphere. When the surface blocked by the object is equal to $1m^2$, we say that the solid angle corresponding to this object is $1sr$.($\Omega = \frac{area}{r^2} = \frac{1m^2}{1^2m^2} = 1sr$) .

Another example: an object that covers the whole sphere will have a solid angle of $\Omega = \frac{area}{r^2} = \frac{4\pi r^2}{r^2} = \frac{4\pi 1^2 m^2}{1^2 m^2} = 4\pi$

Important note: the area used in the calculations of the solid angle is not the "face" area of the object, but the area on the sphere that is blocked by the object. There are cases when we can approximate these two as being equal, but this is generally not the case.

5.2.2 The solid angle subtended by a cylinder

In our experiment, the detector has a cylindrical shape, so it makes sense to discuss how the solid angle changes as we change the distance between the detector and the source. Figure 5.1 shows a diagram of how the detector is seen from the source. According to the definition

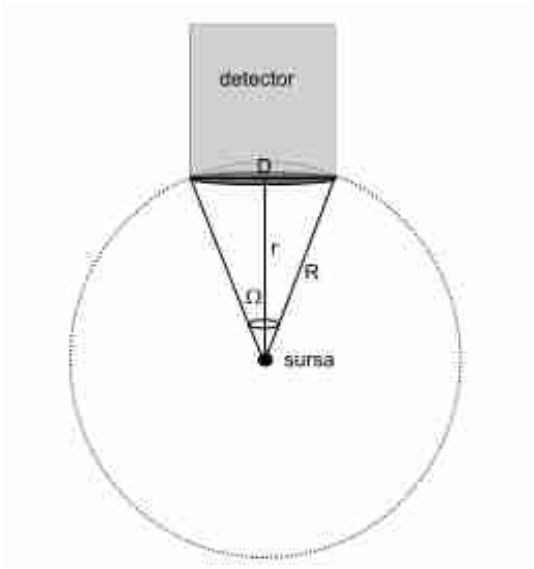


Figure 5.1: Diagram showing the solid angle subtended by the detector

of the solid angle

$$\Omega = \frac{\text{Spherical cap area}}{R^2} = \frac{2\pi R(R-r)}{R^2}$$

where we used the formula for the area of the spherical cap.

Introducing $R^2 = (D/2)^2 + r^2$ in the relation we obtain

$$\frac{\Omega}{4\pi} = \frac{1}{2} - \frac{1}{\sqrt{4 + (D/r)^2}} \approx \frac{D^2}{16r^2} \tag{5.2}$$

where D is the diameter of the detector face and r is the distance between the source and the detector.

Using equation 5.2, equation 5.1 can be written as:

$$n = \frac{N}{\Delta t} = \Lambda \cdot \frac{D^2}{16r^2} \cdot \epsilon \cdot S \tag{5.3}$$

5.3 Description of the experiment

The experiment uses the following items:

- scintillation detector, equiped with photomultiplier tube; the detection efficiency of the detector is 0.2;
- digital counter, model NP 632;
- rack for positioning the source at different distances from the detector;
- plastic tray that will hold the radioactive source ;
- lead shield;
- radioactive source (^{137}Cs) (you need to ask the assisting processor to bring you the source).

5.4 Work procedure

1. Identify the components of the experimental setup.
2. Perform a measurement of the background (600 seconds long).
3. Measure the distances at which the source will be placed.
4. Ask for the source.
5. Place the source on the tray and position the tray in one of the slots.
6. Put the lead shielding in place.
7. Perform a measurement of the count rate (using a measurement time of 60-100 seconds).
8. Repeat steps 5-7 for all the different available positions of the source tray.

Exmple of data table:

r(...)	t(...)	N(counts)	$\sigma_N(\dots)$	n'(...)	$\sigma_{n'}(\dots)$	$n(\dots)$	$\sigma_{\dots}(\dots)$
background							
3							
4							

5.5 Data analysis

The steps of the analysis are the following:

1. Answer questions 3, 4.
2. Make the graph $count\ rate = f(distance)$. Estimate from it the source activity, using the answer to question 4.
3. Make the graph $count\ rate = f(solid\ angle \approx \frac{D^2}{16r^2})$ and, using formula 5.3, estimate the source activity.

Use the values of $\epsilon = 0.2$, $D = 2.54\ cm$ and $S = 1$.

5.6 Uncertainty calculations

Calculate the uncertainty of the activity from the uncertainty of the slope using the least square method.

5.7 Conclusions

Method	Source activity (...)	Uncertainty (...)
I		
II		

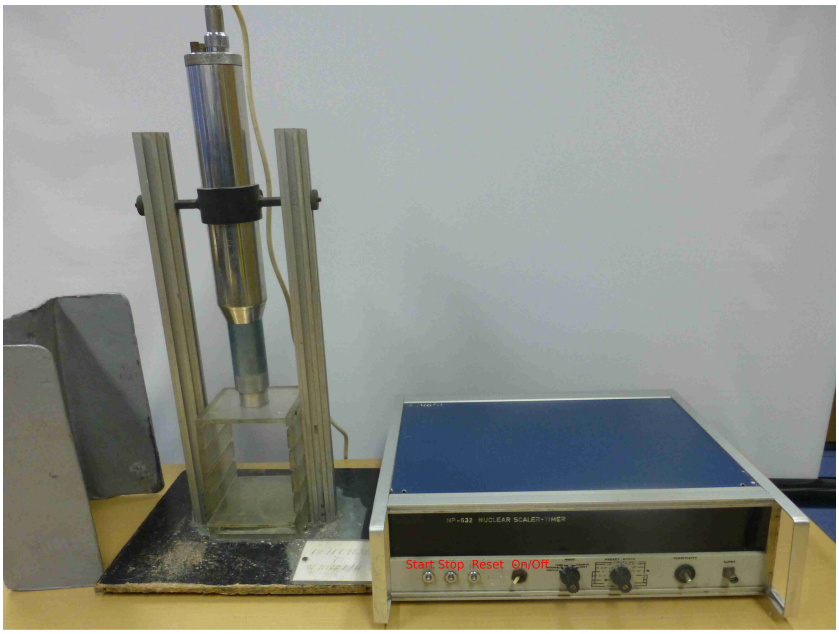


Figure 5.2: Photo of the experimental setup

5.8 Questions

1. Can this method be used to measure the activity of beta- or alpha-radioactive sources?
2. How does the solid angle of an object change when we double the distance between the apex and the object?
3. Why is a measurement of the background necessary? How is the background influencing the measurement result and how can we correct for it?
4. What would the value of the solid angle be if the source could be placed at distance zero from the detector?
5. What is the unit, in SI, for the activity?
6. Demonstrate the second part of equation 5.2. What assumption was made to obtain it? Is this assumption valid for all measurements?
7. How many significant digits should the uncertainty be written with? How many significant digits should the final result be written with?

Note: The official document describing the processing of measurement data and their uncertainties is "Evaluation of measurement data - Guide to the expression of uncertainty in measurement" published by BIPM, JCGM 100:2008. (You can find this document by for "GUM BIPM"). Section 7.2.6 (reproduced bellow) defines the number of digits to be used in expressing the uncertainty in measurements. **For the purpose of our measurements, we can use a single significant digit (or two if that significant digit is 1).**

" 7.2.6 The numerical values of the estimate y and its standard uncertainty $u_c(y)$ or expanded uncertainty U should not be given with an excessive number of digits. It usually suffices to quote $u_c(y)$ and U [as well as the standard uncertainties $u(x_i)$ of the input estimates x_i] to at most two significant digits, although in some cases it may be necessary to retain additional digits to avoid round-off errors in subsequent calculations.

In reporting final results, it may sometimes be appropriate to round uncertainties up rather than to the nearest digit. For example, $u_c(y) = 10,47$ mΩ might be rounded up to 11 mΩ. However, common sense should prevail and a value such as $u(x_i) = 28,05$ kHz should be rounded down to 28 kHz. Output and input estimates should be rounded to be consistent with their uncertainties; for example, if y = 10,05762 Ω with $u_c(y) = 27m\Omega$, y should be rounded to 10,058 Ω. Correlation coefficients should be given with three-digit accuracy if their absolute values are near unity. ”

Experiment 6

Determination of the attenuation coefficient of gamma radiation in matter

6.1 Purpose of the experiment

- Study of the attenuation of gamma radiation by different materials;
- Determination of the mass attenuation coefficient for three materials.

6.2 Theoretical context of the experiment

When a nucleus changes its internal structure (a process called nuclear decay), it can emit various particles: neutrons, electrons, protons, alpha particles, photons, etc. High energy photons are also emitted when high energy electrons are interacting with a solid target. For historical reasons, the photons emitted in nuclear processes are named *gamma photons*, while those emitted in atomic processes are called *x-rays*. These photons have much higher energy than photons in the visible spectrum. Due to their high energy, they interact weakly with matter. The probability of the photon interaction with a certain material depends on the photon energy and on various characteristics of the materials:

- density of the material. For example, the photons will interact more with solid water than with water vapor.
- atomic number of materials. For example, photons will interact more with Lead ($Z = 82$) than with Aluminium ($Z = 13$)

This variability makes high energy photons useful in many applications: in medical radiographies we use the fact that bone tissue has much higher density than flesh; we can find cracks in thick solid objects because the density of the crack (seen as space inside the solid filled with air/gas) is much lower than that of the surrounding medium.

The equation that describes how the intensity of the radiation decreases as it traverses a material slab is called **Lambert's Law**. This variation can be explained using a relatively simple model. Imagine a monoenergetic photon beam (gamma or x-rays) of intensity I_0 arriving under normal incidence at the surface of a material slab as depicted in Figure 6.2. As the beam goes through the material its intensity is continuously reduced. Consider that after reaching a depth x the intensity of the beam is $I(x)$. After traversing another very

thin slice of thickness dx, the photon beam intensity is reduced by another small amount dI. If we make the assumption that the reduction dI depends on the intensity of the photon beam $I(x)$, the properties of the material (denoted by the constant μ, named *attenuation coefficient*) and the thickness of the material dx, we can find a mathematical expression for the way the intensity changes inside the material:

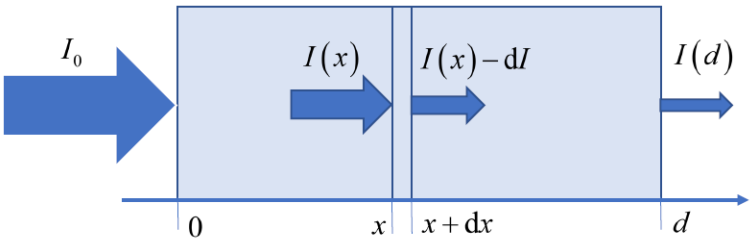


Figure 6.1: Attenuation of the photon flux inside a thick slab of material.

$$-dI = \mu I(x)dx \tag{6.1}$$

By integrating this relation we obtain the law of variation of the intensity as a function of the material thickness (**Lambert's Law**) :

$$I(x) = I_0 \cdot e^{-\mu x} \tag{6.2}$$

where I_0 is the incident beam intensity, $I(x)$ is the intensity of the the beam after traversing a thickness x and μ is the *attenuation coefficient*.

In practice, the values of the attenuation coefficient are experimentally measured. They can be found on the "XCOM:Photon Cross Sections Database" maintained by NIST (National Institute of Standards and Technology - USA). The important experimental observation is the fact that the attenuation coefficient for a given material has different values for different energies of the incoming photons.

6.3 Description of the experiment

The experiment uses the following items:

- scintillation detector, with photomultiplier tube;
- computer, connected to the detector;
- radioactive source (^{137}Cs) (you need to ask the assisting professor to place the source in the experimental position);
- three stacks of different materials;

6.3.1 Details on how to start the measurement

- After everything is turned on, the high voltage on the detector should be 0.67 kV, as seen in figure 6.2.

- On the PC, lauch the application named "CASSYlab". A "Settings" window will appear. Activate the measuring channel by pressing the red button, which opens another window named "Measuring parameters".
- Move the slider on the "Gain Box A1" to a value around "-5.81".
- Close the two windows.
- Start the measurement by pressing the stopwatch-icon button; the time of the measurement is shown in the upper-right corner.

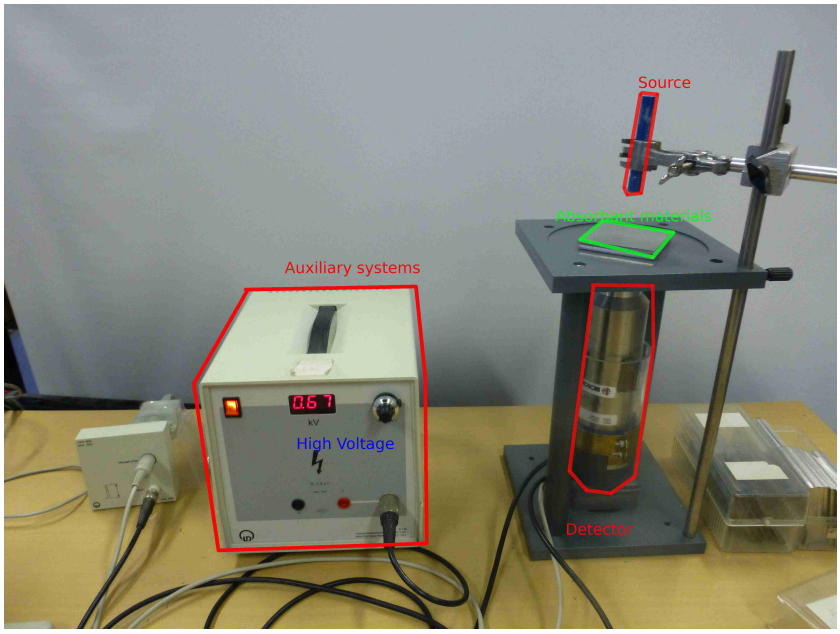


Figure 6.2: Experimental setup for the measurement of the attenuation coefficient

6.3.2 Calculating the intensity of the photon beam

The detector records information about each of the photons that interacts with it. Not all the photons deposit all their energy in the detector. The information about the photons that interacted with the detector is presented as a spectrum. A spectrum is a graph showing how many times the photons deposited a certain amount of energy, $Number\ of\ counts = N(Deposited\ energy)$.

After two measurements with different absorber thicknesses, the display of the computer will look similar to figure 6.3. The number of counts we are interested to measure are those in the "photopeak" of the spectrum, as marked in the figure. The two spectra are taken at two different thickness values of the absorber, and it can be seen that the number of counts is different for the two measurements (represented with two different colors). The real shape of the area of interest is given by a Gauss curve, but for the purpose of our experiment it can be approximated with a triangle.

The photopeak area is proportional to the photon beam intensity reaching the detector and with the time of measurement. If the spectra are acquired in different time intervals, the corresponding areas must also be divided by the duration of measurement.

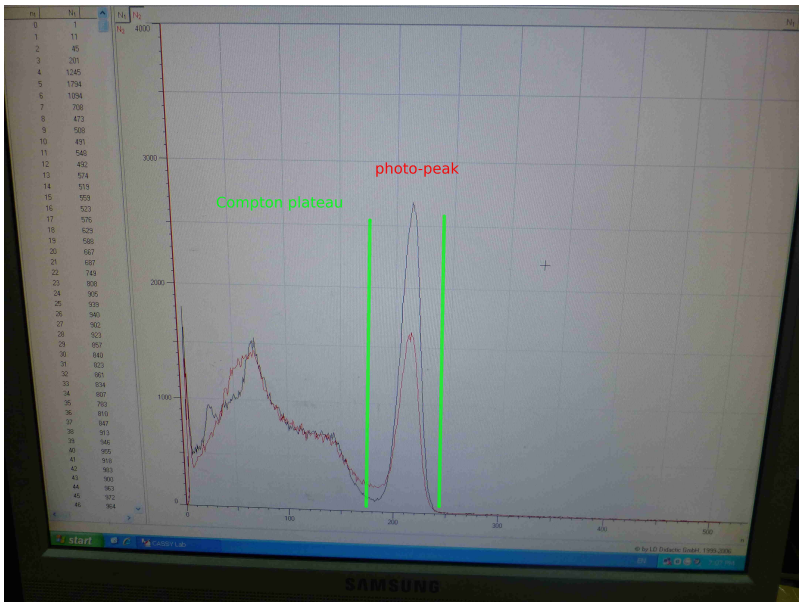


Figure 6.3: Experimental spectra obtained using the ^{137}Cs source for two different thicknesses of the absorber.

6.4 Work procedure

1. Identify the elements of the experiment;
2. Turn on the computer and the detector using the instruction in section 6.3.1;
3. (optional) Perform a measurement of the background spectrum;
4. Perform a measurement with the source without any material between the source and the detector (ask the assisting professor to place the source in the experimental position);
5. Place absorbant sheets beween the source and the face of the detector; write down the material and the thickness;
6. Perform a measurement, calculating from the obtained spectrum a measure of the number of photons reaching the detector; note the measurement duration; (you can follow the procedure in section 6.3.2)
7. Making sure you are not mixing the materials, repeat step 5 - 6 for different combinations of thicknesses . The largest thickness used MUST insure that the count rate decreases by at least a factor close to e;
8. Repeat steps 5-7 for the other two materials.

Example of table(one for each material):

No	Thickness(...)	Height (...)	Left channel(...)	Right channel(...)	Area (...)	ln ...
1						
2						
3						
4						
⋮						

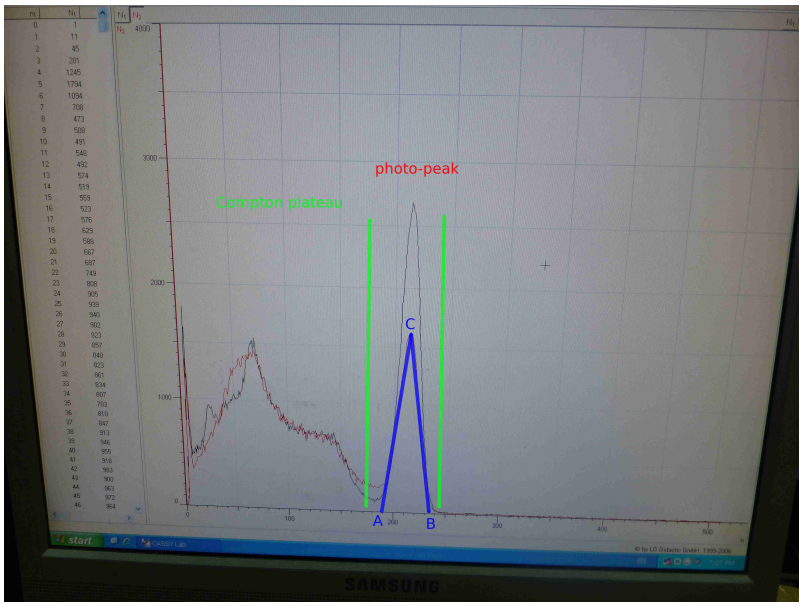


Figure 6.4: Example of how to measure the area of the photopeak.

6.5 Data analysis

1. Answer question 1;
2. Using the tables obtained previosly, make the graphical representation that allows you to calculate the coefficient of attenuation;
3. Calculate the coefficient of attenuation from the analysis of the graph;
4. Calculate the coefficient of attenuation using the least square method;

6.6 Uncertainty calculations

Calculate the uncertainty of the attenuation coefficient using the uncertainty of the slope calculated using the least square method.

6.7 Conclusions

Material	Measured attenuation coefficient(μ)(...)	Uncertainty (...)	Atomic Number (Z)	Density(...)

6.8 Questions

1. Starting from relation 6.2, write a linear relation between the intensity of the radiation and the thickness of material.
2. What is the uncertainty of the foil thickness. Why?

3. Which of the three materials is the most useful for creating a shield to protect ourselves from gamma radiation? Why? How thick should such a shield be to reduce the incoming radiation by a factor of 1000?

4. In a shielding application (where a certain material is placed between us and the source) what is better: to use a single piece of thickness x or two pieces of thickness $x/3$ and $2x/3$? Why?

5. Knowing that the materials you are using as absorbers are Fe, Pb and Al, how do you identify them?

6. Do you see any difference between the attenuation coefficients calculated with the graphical method vs using the least square method? If you do, is this difference significant?

Experiment 7

Study of the photoelectric effect

7.1 Purpose of the experiment

- Study of the photoelectric effect
- Determination of Planck's constant numerical value

7.2 Theoretical context of the experiment

The photoelectric effect is one of the most important phenomena in physics. First discovered by Hertz in 1887, it was finally explained by A.Einstein in 1905, and his explanation led to the discovery of quantum mechanics.

The photoelectric effects[1] occur when light (made up of particles we call "photons") interacts with electrons that are part of the electronic shell of an atom. The photon is absorbed, its energy being used to release the electron from its bound with the atom. As the electron is bound inside the atom, a certain amount of energy needs to be used to eject it, the remaining energy being transformed into kinetic energy of the electron. The fact that some energy is needed to eject the electron explains why the photoelectric effects occurs only when the energy of the photon is higher than this threshold value, which we call the "extraction work", in a parallel with classical mechanics where the energy and work are inextricably related. The photoelectric effect can be then described by the following energy conservation relation:

$$E_{photon} = W_{extraction} + K_{electron} \tag{7.1}$$

where:

- E_{photon} is the energy of the incoming photon (related to its frequency by $E_{photon} = h\nu$, with h being known as Plank's constant)
- $W_{extraction}$ is the energy required to free the electron from its atomic bound
- $K_{electron}$ is the kinetic energy of the electron

We can write the relation as:

$$K_{electron} = h\nu - W_{extraction} \tag{7.2}$$

As a consequence, Planck's constant can be calculated from measurements of the kinetic energy of the electron emitted after photoelectric absorption of monochromatic photons.

[1] In this document we speak only of the external photoelectric effect.

7.3 Description of the experiment

The experiment uses the following items:

- light source, producing photons with a number of discrete energies. (In this experiment we are using a Hg light source (see appendix 7.3.2)). To select a single energy for the incoming photons a rotating wheel cotaining different filters is placed after the source. Each of the filters in the wheel will select light with a color (frequency) centered on a certain value.
- optical equipment: a diafragm placed after the source will allow the variation of the intesity of light; a lens will concentrate the light coming from the source on the photocell
- photocell: a photosensitive element, part of an simple electric circuit.

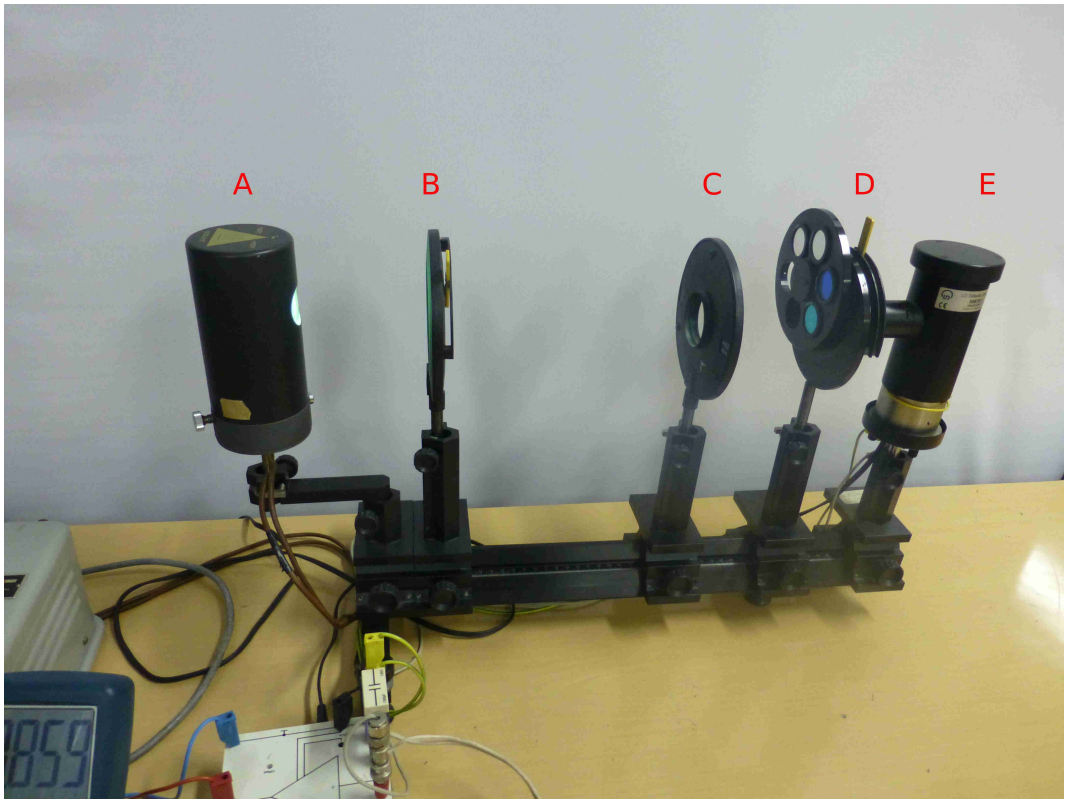


Figure 7.1: Photograph of the experimental setup

7.3.1 Measuring the kinetic energy of the electron

The kinetic energy of the electron is measured using the simple circuit shown in figure 7.2.

In this circuit the photosensitive element (named K on the diagram) is placed inside a small air-free enclosure. Following the iradiation, the electrons are being extracted (emitted) by the cathode with a certain kinetic energy given by relation 7.2. The first electrons travel freely through the vacuum and land on the anode A. In this respect, the combination of cathode + light + anode is similar to a current source. As more electrons travel from the cathode (K) to the anode (A), the potential difference between the cathode and anode increases. This process, the motion of charged particles in electric fields is described in detail during the course. As discussed there, if we denote by U the potential difference, then only

the electrons with kinetic energy higher than the value eU can reach the anode. The process ends when, at a certain value of the potential difference, which we denote by U_0, none of the electrons emitted by the cathode have enough kinetic energy to travel to the anode. Therefore, the value of the potential difference is a direct measure of the kinetic energy of the photoelectrons.

The external circuit is made of a single capacitor which gets charged as electrons are travelling between the cathode and the anode. The capacitor is connected to a voltmeter and (not shown on the diagram) an external switch that discharges the capacitor, setting the potential difference between its plates to 0V.

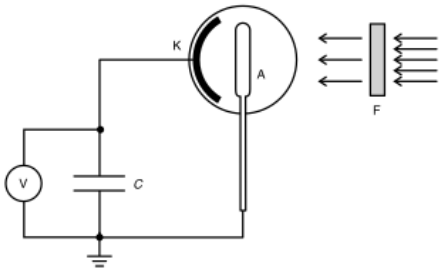


Figure 7.2: Diagram of the electric circuit

7.3.2 Spectrum of Hg lamp

Please see *https://en.wikipedia.org/wiki/Mercury-vapor_lamp*

The strongest peaks of the emission line spectrum of a high-pressure mercury lamp are:

Wavelength(nm)	Color
184.45	ultraviolet
253.7	ultraviolet
365.4	ultraviolet
404.7	violet
435.8	blue
546.1	green
578.2	yellow-orange

7.4 Work procedure

The steps of the experiment are the following:

1. identify the elements of the circuit and the external switch;
2. explain why the equation that relates Planck's constant(h) to the voltage reading and the photon wavelength is:
$$eU_0 = h\nu - W_{extraction} \tag{7.3}$$
3. turn on the multimeter and set it up as a voltmeter;
4. turn on the mercury lamp (it needs 5'-10' to properly warm up);
5. select one of the filters;

6. discharge the capacitor by using the external switch;

7. wait until the capacitor is charged and read the voltage. Write the experimental result in a table (see the example bellow);

8. repeat steps 6, 7 until you have the required number of measurements (as the voltage varies in time, you can also make readings at 15-20" intervals without always discharging the capacitor);

9. repeat steps 5-8 for the other filters available.

Example of table:

	Filter	wavelength (...)	Voltage(...)	$\overline{U_0}$ (...)	$\sigma_{\overline{U_0}}$(...)
1	yellow				
2			⋮		
...					
10					
1	green				
2			⋮		
...					
10					
...	...	...	...	...	...

7.5 Data analysis

- For each of the filters, we can assume that the measurements were performed in identical conditions, hence the maximum energy of the photoelectrons is related to $\overline{U_0}$, obtained by calculating the average of the measurements with the same filter. Calculate the average value $\overline{U_0}$ and add it to the table; also, calculated the standard deviation of the mean ($\sigma_{\overline{U_0}}$) and add it to the table.

- On graph paper represent U_0 as a function of incoming light frequency (or 1/wavelength).

- Calculate the value of Planck's constant from the analysis of the graph.

- Calculate the value of Planck's constant using the least square method.

7.6 Uncertainty calculations

Calculate the uncertainty of Planck's constant from the uncertainty of the slope obtained using the least square method.

7.7 Conclusions

Planck constant(h)(...)	Uncertainty (...)

7.8 Questions

1. The relation between the kinetic energy of the photoelectrons and the frecquency of the incoming photons is given by what relation? What type of dependence exists between the the kinetic energy of the photoelectrons and the frequency of the radiation?

2. In figure 7.1 the purpose of the elements is the following:

A	
B	
C	
D	

3. If 10 measurements of the same variable x_i are made in identical conditions, what is the average value? Write the relation for the standard deviation of the data. Write the relation for the standard deviation of the mean.

4. what is the eV (electronvolt)? What is its value in the international system of units?

5. What is the kinetic energy and velocity (as a fraction of c-speed of light) of an electron starting from rest and being accelerated under a potential difference of $10V$? What if the potential difference is $100V$? What if the potential difference is 10^8V?

6. From your data analysis, what is the lowest frequency - ν_{min} or (longest wavelength λ_{max}) for which the fotoelectric effect can occur? What is the significance of this value?

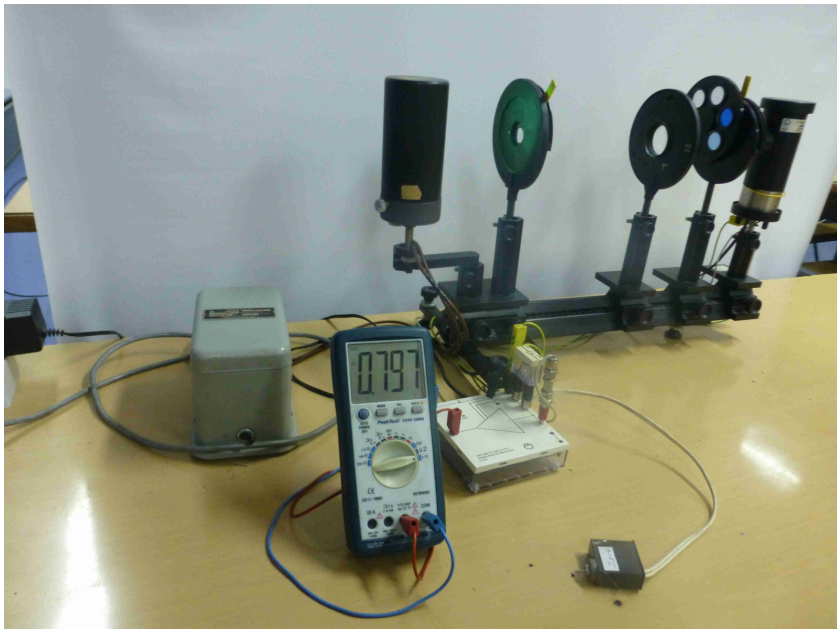


Figure 7.3: Photo of the experimental set-up

Experiment 8

Study of the Hall effect

8.1 Purpose of the experiment

- experimental measurement of the Hall constant
- experimental measurement of the charged carrier concentration in a semiconductor

8.2 Theoretical context of the experiment

The Hall effect is a phenomenon that occurs when a thin conductor or a semiconductor strip carrying an electric current is placed in a magnetic field perpendicularly to the current flow. It was discovered in 1879 by E. H. Hall and it consists in the occurrence of an electric field called the Hall field E_H , whose field lines are at right angles both with the flow direction and the magnetic field lines.

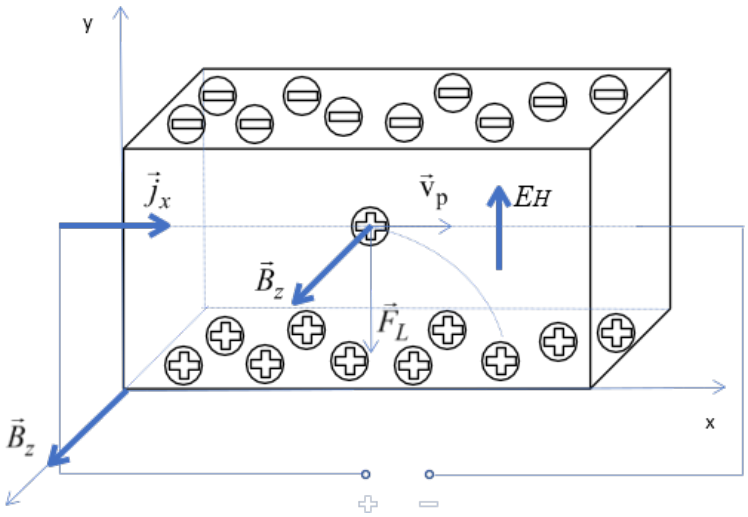


Figure 8.1: Hall effect in a p-type semiconductor

The effect is due to the spatial separation of the free charge carriers moving in the magnetic field. A voltage (the Hall voltage) is developed across the strip perpendicularly to the direction of the current and that of the magnetic field.

The transport of charge carriers in solid materials can be described in classical physics by taking into consideration both their random collective motion and an ordered motion in the direction of the external electric field. The charge carrier trajectories are deflected by the Lorentz force $\vec{F_L} = \pm e\vec{v_x} \times \vec{B}$ toward the lower side (see Figure 8.1). In the following we will discuss the effect in semiconductors. In Figure 8.1, a simplified representation of the Hall effect in p-type semiconductors (semiconductors in which the the majority charge carriers are positive (holes)) is given.

By applying an external voltage in the Ox direction, an electric current I is set up in the p-type semiconductor. In Figure 8.1 the current density vector $\vec{j_x}$ is represented in the positive x-direction. The current density is defined as the electric current per unit transverse area. Obviously, the electrons will move in the negative x-direction.

Let us consider that a magnetic field of induction $\vec{B}$ is applied in the positive z-direction. Then, the positive charge carriers (holes) in the flow will be deflected by the Lorentz force $\vec{F_L} = +e\vec{v_x} \times \vec{B}$ toward the lower side of the p-type semiconductor, as shown in Figure 8.1. Thus, an electric field $\vec{E_H}$, called the Hall electric field, appears in the positive y direction and, consequently, a Hall voltage U_H can be measured across the strip, in this direction.

The motion of the charge carriers in the direction of the Hall field is characterized by an average transport velocity which is called the drift velocity. It is noted by v_p for the holes (positive carriers). The average transport speed acquired by carriers per unit Hall electric field is called mobility. The mobility of the positive charge carriers (holes), defined by

$$\mu_p = \frac{v_p}{E_H}. \tag{8.1}$$

is generally smaller than the mobility of the electrons, defined by

$$\mu_n = \frac{v_n}{E_H} \tag{8.2}$$

The mobility is a positive number which quantifies the effect of an electric field on the motion of the charge carriers, in terms of the speed change. It is an important parameter for the functioning of semiconductor devices.

Note that the motion of the charge carriers in the semiconductor strip has two components: a motion in the direction of the external electric field (Ox) and another along the Hall field lines (Oy), due to Lorentz force. The resultant force acting on the majority positive charges is equal to the vector sum between the electric force $\vec{F_x} = e\vec{E}$ and the Lorentz force $\vec{F_L} = +e\vec{v_x} \times \vec{B}$.

$$\vec{F} = \vec{F_x} + \vec{F_L} = +e\vec{E} + e\vec{v_x} \times \vec{B} \tag{8.3}$$

When equilibrium is reached along the Hall field lines, the corresponding net force becomes zero. The scalar equation expressing the force equilibrium in the y direction reads

$$e \cdot E_H = e \cdot v_x \cdot B \tag{8.4}$$

and thus

$$E_H = v_x \cdot B. \tag{8.5}$$

From this condition 8.5 we obtained that the Hall electric field intensity is proportional to the component of the speed along the external field and to the magnitude of the magnetic field induction.

To find the speed v_x in terms of the electric current I injected in the semiconductor strip, we need the sizes of the rectangular parallelipiped which is a model for the semiconductor

strip. The dimensions in the x, y, z directions will be noted by a, b, and c. The local form of Ohm's law gives the relation between the current density j_x, the electric charge density and the speed v_x:

$$j_x = +e \cdot p \cdot v_x, \tag{8.6}$$

In 8.6 we used the concentration of positive charge carriers p (holes) to calculate the electric charge volume density $+ep$. The hole concentration is the number of holes per unit volume. Then, the speed v_x is

$$v_x = \frac{j_x}{ep} \tag{8.7}$$

The previous equation can be written in the form

$$v_x = R_H j_x \tag{8.8}$$

by using the notation

$$R_H = \frac{1}{+ep} \tag{8.9}$$

R_H is the Hall constant. It is the inverse of the charge volume density of the semiconductor. The curent density in the simplified geometry can be calculated as

$$j_x = \frac{I}{bc}, \tag{8.10}$$

where bc is the area transverse to the flow of charges. Using 8.10 in 8.8, we get the speed v_x as a function of the electric current I injected in the semiconductor strip.

$$v_x = R_H \frac{I}{bc}. \tag{8.11}$$

Then, acording to 8.11 and 8.5, the Hall electric field intensity is related to the electric current intensity I injected in the Ox direction and the magnetic field induction B applied in the Oz direction through the equation

$$E_H = R_H \cdot \frac{I}{bc} \cdot B = R_H \cdot \frac{I}{bc} \cdot B \tag{8.12}$$

and the corresponding Hall voltage U_H, which is obtained in the Oy direction, is

$$U_H = E_H \cdot b = \frac{R_H}{c} IB. \tag{8.13}$$

8.3 Description of the experiment

The experimental setup is shown in figure 8.2. The studied semiconductor is already placed inside the Hall-effect-module.The module is powered by the 12V AC output of the power unit over the ac-input on the backside of the module. The injected current I and the ambiental temperature can be read on the integrated display of the module. Its value can be adjusted using the knob bellow the display.

The magnetic field is produced by two series-connected coils fed from the DC outlet of the main supply unit. It is advisable for this purpose to set the voltage to the maximum value and to adjust the magnetic field to the desired value by use of the current control knob.The magnetic induction of the field is measured by the teslameter, the Hall probe of which is placed at the centre of the field (after the apparatus has been adjusted). The Hall voltage is measured by the high-resistance digital multimeter.

The dimensions of the semiconductor piece are the following:

$$a = 20mm$$

$$b = 10mm$$

$$c = 1mm$$

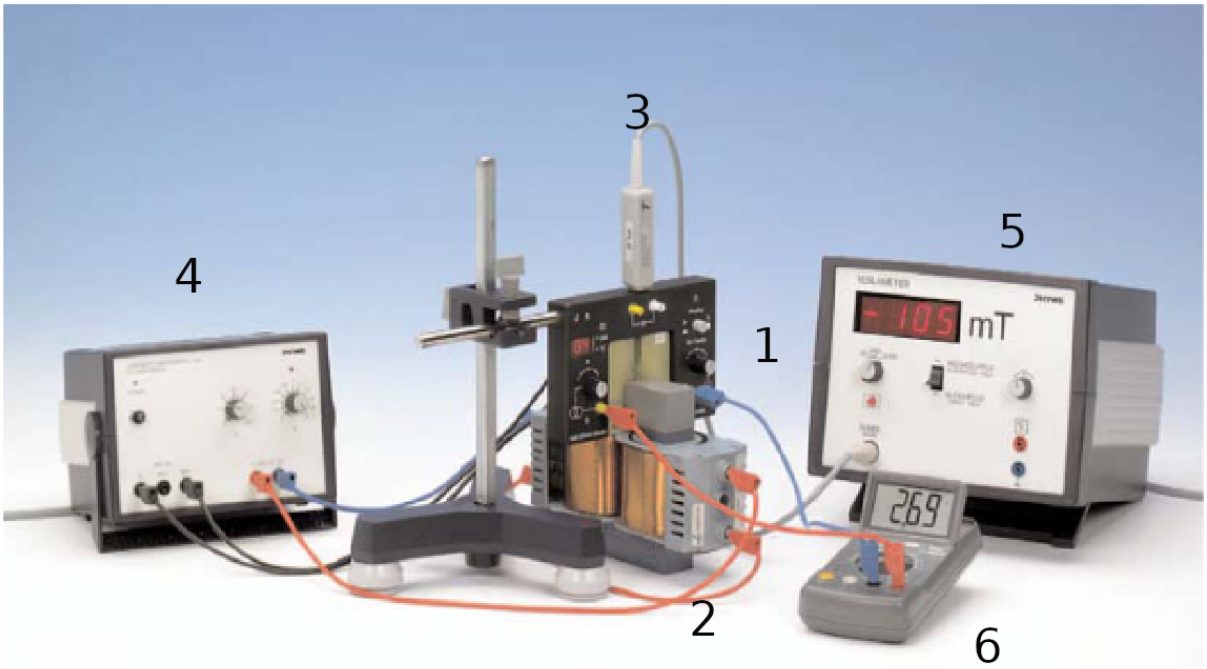


Figure 8.2: Photograph of the experimental setup

The components of the experimental setup are as follows:

1. Hall effect module with semiconductor plate
2. Coil, 600 turns, with Iron core and pole pieces
3. Hall probe
4. Power supply 0-12 V DC, 12 V AC
5. Teslameter, digital
6. Digital multimeter

8.4 Work procedure

The steps of the experiment are the following:

1. identify the elements of the experiment
2. place the Hall probe in the position shown in the figure, and measure the field at the probe location. When no current is flowing through the magnetic coils, tune the magnetic field dial such that the magnetic field is 0 mT.

3. using the push-button on the module read the ambient temperature and then choose the setting that shows the intensity of the current flowing through the probe.

4. connect the multimeter to the sockets of the Hall voltage U_H on the front-side of the module

5. set the current flowing through the probe to 10 mA using the control knob.

6. change the magnetic induction from -300 mT to 300 mT in steps of 50 mT. (You can change the polarity of the magnetic field by swaping the polarity of the DC current generating the magnetic field)

7. record the values of the Hall voltage for each value of the magnetic field

8. repeat steps 5-7 for currents of 20 mA and 30 mA

Example of table:

B(mT)	$U_H(mV)$ $(I = 10mA)$	$U_H(mV)$ $(I = 20mA)$	$U_H(mV)$ $(I = 30mA)$
-300 mT			
-250 mT			
...	...	...	...

8.5 Data analysis

- answer question 4
- on graph paper represent U_H as a function of the induction of the external magnetic field. (note: you may draw all the experimental values on the same graph, making sure you mark them with different symbols)
- calculate the values of the Hall constant from the analysis of the graphs.
- calculate the values of the Hall constant using the least square method on one of the data sets. (note: if possible, each group member use a different set)
- calculate the concentration of the positive charges (holes) from the value of the Hall constant obtained using the least square method.

8.6 Uncertainty calculations

Calculate the uncertainty on the value of the Hall constant from the uncertainty of the slope obtained using the least square method.

Calculate the uncertainty of the value of the concentration of positive charges (holes).

8.7 Conclusions

Concentration of holes (...)	Uncertainty (...)

8.8 Questions

1. Draw the forces acting on an electron moving in the opposite direction of the positive charge carrier (hole) in figure 8.1.

2. What happens with the sign of the Hall voltage if the current is carried inside the semiconductor by electrons instead of holes (n-type semiconductor instead of a p-type semiconductor)?

3. Draw and justify the trajectory of a positively charged particle moving in region of uniform magnetic field perpendicular to its velocity.

4. What is the relation between the value of the Hall voltage, the induction of the external magnetiv field and the intensity of the current flowing through the probe? What type of relation exists between the Hall voltage and the external magnetic field induction for a given value of the current flowing through the probe?

5. What is the uncertainty propagation formula? Using the uncertainty propagation formula calculate the uncertainty of function $f(x) = \frac{1}{a \cdot x}$ where a is a constant and x is an experimental value known with an uncertainty σ_x.

6. What is the uncertainty of the experimental values measured during the experiment? Why?

,

Appendix A

Uncertainty in counting experiments

A.1 Definitions

Assume you are measuring any quantity, which we'll denote x and, for the sake of simplicity assume that the measurements can only take integer values.

Assume that you have taken N measurements $(x_1, x_2, \ldots, x_N)$, where each of these is a numerical value (for example, $x_3 = 2cm$). Two elementary properties of this data set are:

- *Sum*: $\Sigma = \sum_{i=1}^{N} x_i$
- *Experimental mean*: $\bar{x}_e = \frac{\Sigma}{N}$ **Note:** The experimental mean is generally different from the mean we would get if we repeat the measurement an infinity of times. We call this ideal mean *true mean* value ($\bar{x}$).

Another way to represent the data is to express it in terms of its *frequency distribution function* $F(x)$:

$$F(x) = \frac{\text{number of times value } x \text{ appears in the data}}{\text{number of measurements (=N)}}$$

From this definition, it becomes clear that

$$\sum_{x=0}^{\infty} F(x) = 1$$

and that

$$\bar{x}_e = \sum_{x=0}^{\infty} x \cdot F(x)$$

Other parameters can be used to caracterise the experimental data:

- *residual* of each experimental point: the amount by which each point differs from the experimental mean value $d_i = x_i - \bar{x}_e$. Note that: $\sum_{i=1}^{N} d_i = 0$.
- *deviation* of the each data point: $\epsilon_i = x_i - \bar{x}$.
- *sample variance*

$$s^2 = \bar{\epsilon^2} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2 \tag{A.1}$$

The sample variance measures how different the measured values are from each other.

Note: for a given set of measurements it is impossible to calculate the sample variance because there is no way for us to know $\bar{x}$. It can be demonstrated that an alternative relation exists, where the true mean is replaced by the experimental mean:

$$s^2 = \frac{1}{N-1} \sum_{i=1} N(x_i - \bar{x}_e)^2$$

Note: The relation above is for sample variance of the data. We can also calculate the variance of the sampling distribution of the mean value

Relation A.1 is the average value of $(x - \bar{x})$ and can be rewritten as:

$$s^2 = \sum_{i=0}^{\infty} (x_i - \bar{x})^2 F(x) \tag{A.2}$$

Conclusions:

- any set of data can be completly described through its frequency distribution function $F(x)$
- two important properties of the frequency distribution function are important: the experimental mean and the sample variance

A.2 Statistical models

In certain conditions we can a priory say which is the distribution function that describes the experimental data.

A.2.1 The binomial distribution

This is the most general statistical model. If we make n trials, with a success probability for each of the trials p, then the probability of counting exactly x successes will be given by:

$$P(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{(n-x)} \tag{A.3}$$

We can demonstrate that

$$\sum_{x=0}^{n} P(x) = 1$$

.

We can calculate:

- the expected average number of successes:

$$\bar{x} = \sum_{x=0}^{n} xP(x) = pn \tag{A.4}$$

- the expected (predicted) variance (which we denote σ^2), as a measure of the predicted scatter around the mean:

$$\sigma^2 = \sum_{x=0}^{n} (x - \bar{x})^2 P(x)$$

Note: we define the "stadard deviation" as the square root of the variance.

For the binomial distribution, we can demonstrate that

$$\sigma^2 = np(1-p) = \bar{x}(1-p)$$

A.2.2 Poisson distribution

The Poisson distribution is a particular case of binomial distribution. It can be deduced from the binomial distribution under the assumption that the probability of success (p) is small. In this case

$$P(x) = \frac{(pn)^x e^{-pn}}{x!} \tag{A.5}$$

This can be (because of the result A.4) rewritten as:

$$P(x) = \frac{(\bar{x})^x e^{-\bar{x}}}{x!} \tag{A.6}$$

For the Poisson distribution we have:

- mean value (same as binomial distribution):

$$\bar{x} = \sum_{x=0}^{n} (x - \bar{x})^2 P(x) = pn$$

- variance:

$$\sigma^2 = \bar{x}$$

so that the predicted standard deviation is

$$\sigma = \sqrt{\bar{x}}$$

A.2.3 The Gaussian or Normal Distribution

In certain cases the Poisson distribution can be reduced to a more common distribution, the Gauss distribution. This approximation is true when the number of events measured is large. Then, while the distribution is discrete, the variations can be considered continuous, and the distribution becomes:

$$P(x) = \frac{1}{\sqrt{2\pi\bar{x}}} exp\left(-\frac{(x-\bar{x})^2}{2\bar{x}}\right) \tag{A.7}$$

For the Gaussian distribution we can calculate its variance as:

$$\sigma = \sqrt{\bar{x}}$$

A.3 Estimation of the precision for a single measurement

In most cases we need to estimate the uncertainty associated with a single measurement (or a reduced number of measurements). The uncertainty of the measurement is related to how different would the results be if we would repeat the measurement a number of times. In statistical terms, our goal is to estimate the sample variance that would characterize the set of measurements from making a single measurement.

First, let's start with an assumption: by repeating the experiment a high number of times we would obtain a series of experimental results that are following either the Poisson or Gaussian distribution. (This assumption is valid in most experimental cases, due to the *Central limit theorem*). If we make a large number of measurements, then we can match

the mean of the distribution to the mean of the data and the variance of the distribution to the distribution of the data. If we make a single measurement, then we have no other choice than to say that the only measurement we have made is the most likely measurement that could have come from our distribution, which is to say that we assign to the mean of the distribution the value we have measured. Because we assumed that the distribution is Gaussian or Poisson, then, by knowing the mean, we also know the predicted variance, which in turn means that we can estimate what the variance of experimental data would be, if we repeat the experiment.

We can then conclude that $\sigma = \sqrt{x}$, which is to say that the best estimate for the deviation from the true mean is related to the single measurement we have performed.

Appendix B

Least squares method for the determination of the linear fit coefficients

Given a set of pairs of data (x_i, y_i), we aim at finding the model parameters that best describe the data. If we assume that a certain functional relation exists $y = f(x, p_1, p_2, ...)$ then we can use the experimental data to find the values of the parameters that best fit the experimental data. For example, if the data are spread along a line, or if we have reasons to believe the dependence is linear, then the goal would be to find the best line that fits the data.

If the parameters of the linear fit are a and b, then for a measured x_i we expect to obtain a value $\hat{y}_i = a * x_i + b$. In our experiment for the value x_i we obtained the experimental value y_i, in general having a different value than the expected value. We call the residual of the measurement the value: $r_i = y_i - \hat{y}_i = y_i - a * x_i - b$.

The principle of the least squares method is to find the values of a and b for which the value of the total sum of squares(S^2) is as small as possible:

$$S^2 = \sum_{i=1}^{n} (r_i)^2 = \sum_{i=1}^{n} (y_i - a * x_i - b)^2$$

This relation is a function of two parameters, hence the condition that it reaches a minimum value translates into the conditions that the partial derivatives about a and b are equal to 0.

$$0 = \frac{\partial S^2}{\partial a} = \sum_{i=1}^{n} 2 \cdot x_i (y_i - a \cdot x_i - b) = 2 \sum_{i=1}^{n} x_i y_i - 2a \sum_{i=1}^{n} x_i^2 - 2 \cdot b \sum_{i=1}^{n} x_i$$

$$0 = \frac{\partial S^2}{\partial b} = -2 \cdot \sum_{i=1}^{n} (y_i - a \cdot x_i - b) = -2 \cdot \sum_{i=1}^{n} y_i + 2 \cdot a \cdot \sum_{i=1}^{n} y_i + 2 \cdot b \sum_{i=1}^{n} 1$$

$$\{ \sum_{i=1}^{n} x_i y_i - a \sum_{i=1}^{n} x_i^2 - b \sum_{i=1}^{n} x_i = 0 \sum_{i=1}^{n} y_i - a \cdot \sum_{i=1}^{n} y_i - b \cdot n = 0 \tag{B.1}$$

This is a system of two equations with a and b as unknows, which can be solved to obtain:

$$a = \frac{n \sum_{i=1}^{n} x_i y_i - (\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n \sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2} \tag{B.2}$$

and

$$b = \frac{(\sum_{i=1}^{n} x_i^2)(\sum_{i=1}^{n} y_i) - (\sum_{i=1}^{n} x_i y_i)(\sum_{i=1}^{n} x_i)}{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2} \tag{B.3}$$

also, it can be shown that:

$$n \cdot b = \sum_{i=1}^{n} y_i - a\sum_{i=1}^{n} x_i$$

The uncertainties of the slope (a) and intercept (b) can be obtained from an uncertainty propagation calculation on their respective relations and we obtain:

$$s_a^2 = \frac{\left(\frac{1}{n-2}\right)\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{B.4}$$

and

$$s_b^2 = \left(\frac{1}{n-2}\right)\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right) \tag{B.5}$$

These relations can be rewritten using the following notations:

$$\begin{aligned} S_x &= \textstyle\sum_{i=1}^{n} x_i \\ S_y &= \textstyle\sum_{i=1}^{n} y_i \\ S_{xx} &= \textstyle\sum_{i=1}^{n} x_i^2 \\ S_{xy} &= \textstyle\sum_{i=1}^{n} x_i y_i \\ S_{yy} &= \textstyle\sum_{i=1}^{n} y_i^2 \end{aligned} \tag{B.6}$$

$$a = \frac{nS_{xy} - S_x S_y}{nS_{xx} - S_x^2} \tag{B.7}$$

$$b = \frac{1}{n}S_y - a\frac{1}{n}S_x \tag{B.8}$$

If we use:

$$s_\epsilon^2 = \frac{1}{n(n-2)}\left[nS_{yy} - S_y^2 - a^2(nS_{xx} - S_x^2)\right] \tag{B.9}$$

then:

$$s_a^2 = \frac{ns_\epsilon^2}{nS_{xx} - S_x^2} \tag{B.10}$$

and

$$s_b^2 = s_a^2\frac{1}{n}S_{xx} \tag{B.11}$$

Appendix C

Interaction of radiation with matter

When a nuclear decay event takes place, one or more particles are emitted by the parent nucleus. Depending on the type of decay, the emitted particles will have very specific characteristics in terms of their type, energies, momentum. By type, these particles can be one of the following:

- photons
- electrons and/or positrons (light-mass charged particles)
- heavy ions
- neutrons

C.1 Interaction of electrons/positrons with matter

Electrons and positrons are light-mass charged particles. As they travel through matter they will continuously feel the electric field generated by the ions and the electrons of the material. Through these continuous interactions, the electrons/positrons will free other electrons from their atomic bounds, a process called ionisation. Through these interactions, the initial electron is continuously losing energy, until his energy is low enough that no other ionisation is possible. Because the mass of the electron is so small, many of these interactions will result in the electron changing their initial direction, leading to broken trajectory inside the material.

The positrons are the antiparticle of the electron, and will behave like a "positive" charge electron until it loses "all" its energy. Once its energy is low enough, it will "meet" one positron and will undergo the process of "anihillation", where their rest mass is transformed two 511keV photons.

C.2 Interaction of heavy charged particles with matter

Heavy charged particles are emitted in alpha-decay processes. Alpha particles are fully ionised He ions. As they travel through the environment, they will interact the same way the electrons do, losing energy while producing electron-ion pairs. The main difference is that the alpha particle are much heavier than the electrons, so their trajectory is mugh harder to change. As a result, in most cases, the trajectory of the alpha particles in matter is a straight line.

C.3 Interaction of photons with matter

When photons have low energy they behave like a wave. In nuclear phenomena, the photons have very high energies (ranging from few eV to many MeV), so their behaviour is particle-like. (This is why in nuclear physics we usually have photons colliding with electrons and atoms.)

When a photon travels through matter it can undergo one of the following phenomena:

1. *photoelectric absorption*: the photon will "collide" with a bound electron at rest. The photon will dissapear and its energy will be used to free the electron from its bound, the remaining energy being transformed in the kinetic energy of the electron. The electron becomes free to move to the material and interact with it.

2. *Compton scattering*: the photon will "collide" with a free electron. The photon is deflected, some of the energy of the photon is taken by the electron. After the collision, a photon (with lower energy than the initial photon) and an electron will be travelling through the material.

3. *Pair production*: a photon of high energy (higher than 1.022 MeV) will generate an electron-positron pair. The energy of the photon is used to create the electron-positron pair, while the remaining energy is taken by the newly created pair as its kinetic energy. The electron and positron will travel through the medium and slowly lose their energy. Once the positron has lost its kinetic energy, it will encounter an electron, and will undergo a phenomena called "anihilation", where their rest mass will be transformed into energy, two 511keV photons being created.

One important aspect in understanding the interaction of the photons with matter is to understand that, for a photon with given energy, all of these processes are possible at the same time. The preference for one or another process to occur is reflected in the idea of cross-section, which is way to quantify the probability for one of the processes to occur. Figure C.1 shows the probabilities for photons in Lead. The figure shows four graphs: the probability of undergoing photoelectric interaction (purple), the probability of Compton scattering (dotted blue line), the probability for pair production (magenta), and the total probability of interaction. The figure shows the interaction probability in lead, but while the absolute values are different for different materials, the main characteristics are the same.

From the graph we can see the following things:

- the total interaction probability decreases as the photon energy increases. (The black curve is lower when the photon energy is higher)

- the photoelectric effect is most probable at low energies (bellow 200 keV in lead). (The purple line is higher than the other two lines when the energy is bellow 200 keV)

- the Compton effect is the most probable for "medium" energies (the dotted blue line is the highest for energies higher than 200 keV and lower than 2MeV).

C.4 The interaction of neutrons with matter

Neutrons are heavy particles that have no charge. As a result, their path through matter is determined exclusively by their physical collisions with the electrons and the ions of the material.

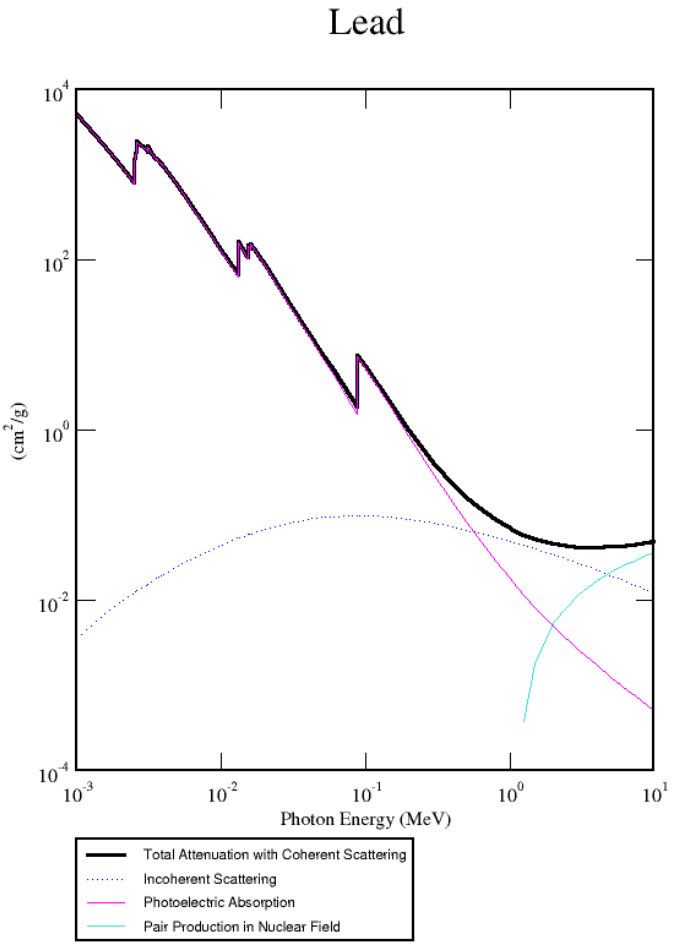


Figure C.1: Photon interaction cross-section in Pb. Please note the logarithmic scale on both axes.

Bibliography

[1]